

ECOLE POLYTECHNIQUE
MSc&T STEEM
Academic Year 2024-25 MANE Vinay

INTERNSHIP REPORT

<< Developing model to determine Operational Expenses for
vehicle fleet on different energies>>

NON-CONFIDENTIAL REPORT

Academic Internship Supervisor: Yvan Bonnassieux, Martin Hennebal
Site Supervisor: Sébastien Le Quoy

Internship dates: Apr-Sept 2025

Annex 2: Declaration of Academic Integrity

Declaration of Academic Integrity

Hereby I, Vinay Mane, confirm that:

1. The results presented in this report are my own work.
2. I am the author of this report
3. I have not used the work of others without clearly acknowledging it, and quotations and paraphrases from any source are clearly indicated.

Vinay Mane
01/08/2025

A handwritten signature in black ink, appearing to read 'V. Mane', with a long, sweeping horizontal line extending to the right.

ANNEX 3 RATING FORM

NAME OF THE STUDENT: **Vinay Mane**

HOST ORGANIZATION: **Vasco Partners**

NAME OF THE SITE SUPERVISOR: **Sébastien Le Quoy**

AUTONOMY	Excellent	<u>Good</u>	Intermediate	Weak
WORK COMMITMENT	<u>Excellent</u>	<u>Good</u>	Intermediate	Weak
INITIATIVE	Excellent	<u>Good</u>	Intermediate	Weak
WORK OUTCOME	Excellent	<u>Good</u>	Intermediate	Weak
GENERAL BEHAVIOR	Excellent	<u>Good</u>	Intermediate	Weak
PROFESSIONAL INTEGRATION	Excellent	<u>Good</u>	Intermediate	Weak
PRESENTATION OF THE WORK	Excellent	<u>Good</u>	Intermediate	Weak

COMMENTS:

We welcomed Vinay for five months to work on a key project for Vasco Partners.

Vinay worked independently on a significant part of the project. However, he also had to integrate with the rest of the team working on other parts of the project, which were closely related to Vinay's work.

Vinay was very committed to his work, particularly during the first part of his internship. Vinay has many good ideas and instincts, particularly in his ability to research the state of the art and various publications related to the work required.

Vinay also demonstrated an ability to develop some of his skills independently in order to meet the requirements of the project (e.g., Python skills). This is very commendable.

However, Vinay needs to develop his ability to work in a team so that he can better integrate the constraints of others into his own work. It is also important to know how to take a step back, keep the main objective in mind, and be able to anticipate deadlines. Finally, it is important to be vigilant about rigor, both in terms of content (e.g., analysis of results) and form (e.g., punctuality).

In summary, we are delighted to have had Vinay during this period.

Work quality: good

Written report: unfortunately, we did not have time to fully review the report as it was written during a vacation period for us. Nevertheless, Vinay has been fully engaged in technical as well as modelization decisions and we have not doubt that Vinay was able to clearly explain the topics he has been working on.

SUPERVISION OF THE INTERN

<u>Everyday</u>	Weekly	Monthly
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Abstract

Fuel costs are a primary expense in the transport sector, and their variability across locations presents a significant opportunity for savings. With development and push towards low carbon fuel, there is a decision for fleet operators to choose and invest in a technology which is cost-effective and feasible for their requirement. We develop a model to compute consumption costs for existing and upcoming energies using Mixed Integer Programming (MIP) and implemented a pipeline to process data and model their usage and constraints. Using MIP, we determine optimal refueling policy for a vehicle with different energies over a study period. The result of this model is the total cost incurred for refueling.

Keywords: road transportation, Total cost of ownership, Consumption, Refueling policy, detour

Acknowledgement

This thesis represents the culmination of several months of dedicated work carried out in collaboration with Vasco Partners. I express my sincere gratitude to Mr. Sébastien Le Quoy, Founder of Vasco Partners, for providing me with the opportunity to work on the GreenFleet project, and for his invaluable guidance, encouragement, and willingness to share his expertise throughout this journey.

The authors thank Pedro Bahia (Federal University of Minas Gerais) for assisting in the formulation of the MILP problem. His unwavering support, insightful feedback, and encouragement over the years have been essential to the successful completion of this thesis work.

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1 INTRODUCTION

1.1 BACKGROUND

The French government provides an online database of fuel prices across refueling stations nationwide [1]. Fuel prices can vary by as much as ~0.2 € per liter (for Diesel) between locations, creating a potential cost-saving opportunity for fleet operators. The spatial variation in fuel prices means there is an opportunity for cost savings if an optimal refueling policy is in place.

The growing adoption of alternative energies—including electric vehicles (EVs), compressed natural gas (GNV), and hydrogen (H₂)—makes it increasingly important to evaluate how spatial variations in energy prices affect refueling strategies. Unlike diesel or biodiesel (B7/B100), where extensive historical data exist, such data for low-emission fuels is sparse. Fueling infrastructure for EVs, GNV, and H₂ remains limited, and relatively few fleets operate these vehicles today, making empirical validation difficult.

Determining an optimal refueling strategy involves more than just selecting stations; the quantity refueled at each stop significantly influences overall trip costs. Literature on refueling optimization highlights the importance of these decisions and considers additional cost and non-cost factors. To enable such optimization, accurate estimation of vehicle energy consumption is required. While simple average consumption rates are sometimes used, they overlook key factors such as gradient, weather, traffic conditions, and vehicle load—all of which substantially affect energy demand. Moreover, fuel consumption has a direct correlation with emissions, making precise modeling critical not only for cost minimization but also for environmental impact assessment.

1.2 BUSINESS REQUIREMENT

Determining Total cost of ownership: Fleet owners sometime do not have an idea as to how much does each vehicle costs in operation. The purchase and rental prices are accounted in the later part of GreenFleet, the biggest component of the Total cost of ownership is Fuel consumption [Michelin white paper] accounting for ~30% of the cost. This is only for B7 and we need to find a way to quantify this for low emission energies.

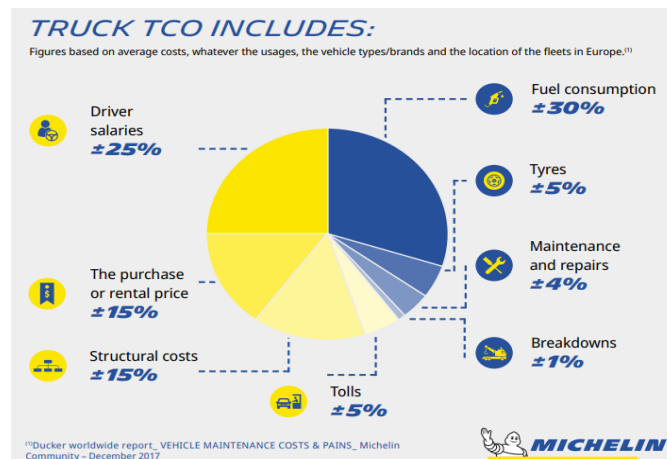


Figure 1: TCO breakdown for a truck fleet in Europe. (Michelin) [2]

Estimating Emission: Heavy Duty vehicles (HDVs) are responsible for more than a quarter of greenhouse gas (GHG) emissions from road transport in the EU and accounts for over 6 percent of the total EU GHG emissions (cite the website climate.ec.europa.eu). As part of the 'Fit for 55' legislative package, the emission standards for HDVs set by Regulation (EU) 2019/1242 [1] were revised. The amended regulation has a wider scope covering nearly all emissions from HDVs as it applies to heavy and medium lorries, city buses, coaches and trailers. The revised targets are more ambitious, aiming for increasing CO₂ emission reductions in the coming decades.

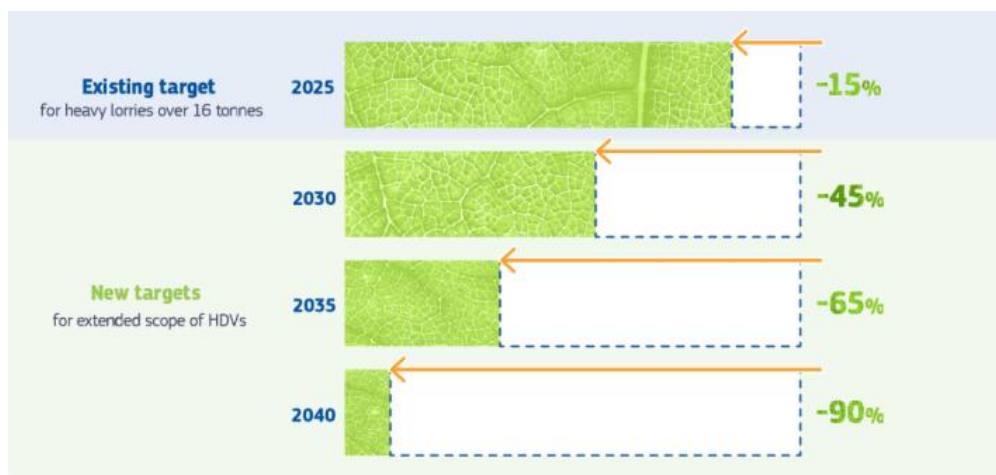


Figure 2: Revised targets for Heavy-duty vehicles emission reduction as part of EU regulation (2019/1242) [3]

Generating a road map for vehicle fleet owners, operators: Upcoming regulations, incentives to transition to low emission fleet and ever increasing prices and risks need to be accounted to forecast future performance and costs for a vehicle fleet. The Greenfleet tool needs to take in operational inputs and prepare a transition plan or a road map for a compliant, cost-effective and resilient fleet for it's customers.

1.3 MOTIVATION

To limit the scope of the report, we focus on the consumption and refueling costs. The total cost of ownership is the sum of CAPEX, OPEX, insurance, subsidies, depreciation, amortization etc. OPEX varies significantly as it depends on how much fuel does a vehicle in a fleet consume, where do we refuel, how much do we refuel. If we can improve the refueling policy, we can reduce the OPEX.

The questions we are trying to answer :

- **Eligibility testing:** If a vehicle operates on GNV, H₂ or any other low carbon fuel, can it operate on the existing route which is performed by a B7 vehicle ?
- **Cost optimization:** Identify the least-cost refueling strategy along each route, accounting for fuel prices, detour distances, driver wages, and refueling times. Can we estimate these costs for low carbon fuels which have not been widely implemented ?
- **Emission estimation:** Provide transparent GHG emission results, broken down into fuel-use and fabrication emissions, using standardized ADEME emission factors and fuel-specific pathways. Can we estimate emissions for vehicles which operate or will operate on low carbon fuel ?
- **Scenario comparison:** Allow operators to compare conventional fuels (e.g., B7) with alternative options (e.g., EV, H₂, GNV, B100, HVO) under different policy or price assumptions. Can we give a comparison and propose which fuel fits the requirement of a vehicle fleet owner/operator ?
- **Scalability:** Process entire transport plans covering hundreds of vehicles and thousands of routes. Can we provide results in reasonable time ?
- **Integration with customer data:** connect to databases containing vehicle catalogs, plan de transport, journey GPS traces, and station datasets. Can we create a tool which can work for many customers and meet their requirement ?

1.4 Problem Statement and Objective

At the core, the problem is a vehicle routing problem (VRP). A Vehicle routing problem is a combinatorial optimization that involves finding an optimal design of routes traveled by a fleet of vehicles to serve certain customers or under given constraint. Traditional VRPs solved for optimal path or fixed path, since we are solving for a fleet of vehicles and the requirement is to replace existing fleet, the problem is taken to be a fixed route VRP (FRVRP).

With environmental and ecological conditions in mind, a new category of VRPs are developed related to green vehicles (G-VRPs)[4]. The approach deals with the challenge of Transportation, Fuel/Energyconsumption and pollution (emissions)

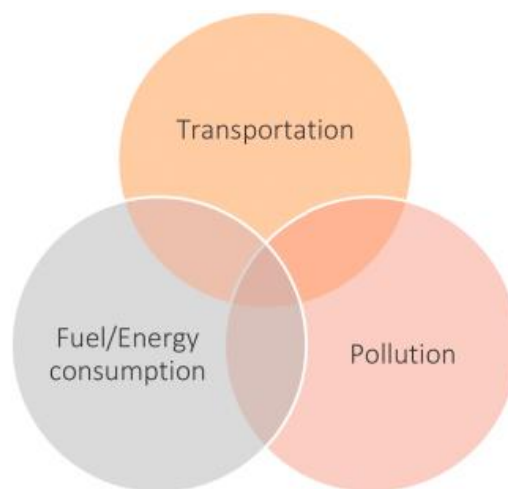


Figure 3 Three structured review term categories of Green VRP[4]

The difference in the approach here are the objectives. Either we solve for minimal energy consumption by path selection, select routes to meet constraints on transportation or have objective to minimize emissions.

Even though FRVRPs have been implemented for Diesel vehicles, this report implements it for low carbon energies because of its simplicity and fits the requirement.

The FRVRP, seeks the optimal refueling policy—i.e., which stations to visit and how much fuel to purchase at each stop—for a given origin–destination route, minimizing total fuel cost. Two broad solution approaches have been explored: heuristic algorithms [5], [6] that produce good feasible solutions within practical timeframes, and exact formulations based on linear or integer programming. In the FRVRP context, the operating cost can be expressed as:

In the FRVRP context, the total operating cost can be expressed as:

$$Cost = \Sigma(\text{refuel_quantity} \cdot \text{price_per_unit_fuel}) + (\text{driver_cost_per_hour} \cdot \text{total_time})$$

We are looking for an exact solution with the objective of minimizing cost. The FRVRP can either be solved as node-based or time_spaced. The difference in the approach is that in node-based we solve over nodes $[i, i + 1]$ and or time dependent example: “Truck driver scheduling with combined planning”[7].

A start and the end of a route is given, these as broken down into smaller journeys and assigned to a vehicle. Geographical positions of refuel stations are mapped in the database and the corresponding price of fuel is known. Driving durations and time constraints are input parameters along with which energy and year the simulation is to be performed. We consider a planning horizon for years given in the input parameter.

The objective is to find an optimal refueling policy, estimate a refueling amount and calculate time-costs associated with refueling with the goal of minimizing it. We limit the cost component to the fuel consumption, refueling and operational hours of the driver.

A vehicle can travel along different routes on different days. For a journey J_1 , we find an optimal refueling strategy for an EV vehicle but the same vehicle couldn't perform J_2 . We introduce a criteria of “**Eligibility**”, if a refueling policy for an fuel type/energy is found, it should be found for all the journey a vehicle makes.

In summary, **the objective is to find Least Cost refueling for all vehicles which are eligible with different energies over the years 2025-2032**

1.5 Report Structure

The report covers Literature review in Section 2, where we mention different approaches to solve the problem.

Section 3 covers methodology, since we don't just solve the problem but create modules for the Greenfleet tool, the overview of system architecture is given. Input, Graph structure and consumption is explained as implemented in the modules. Section 3.2 is the introduction and Section 3.2.1 and 3.2.3 covers 2 approaches to the MILP, 3.2.1 covers approach where a projection is dropped onto a segment for refueling and 3.2.3 directly associates a refuel station to a node. Both the models are same except in the sets and indices (the data structure it needs to solve). The graph structure mentioned in section 3.1.3 is for approach 2.

Section 4 presents overall results and section 5 concludes the report and provides recommendation.

Refer to the appendix for data.

2 LITERATURE REVIEW

In the literature, different refueling problems are analyzed. We limit our attention to FRVRPs and review some reduction techniques for MILP.

In the problem considered by Lin et al. [6] the vehicle traverses a series of gas with different fuel process while traveling along a fixed route (FRVRP). The problem it address is specifically the **“fixed-route vehicle refueling problem”**, with no allowance for detour to find cheaper fuel elsewhere and doesn’t take refueling time into account. The goal is to reach the destination with minimum total fuel costs. This can be achieved by refilling just enough to reach the next cheaper gas station or to fill up the tank if no cheaper gas station is reachable even with a full tank.[6], [7]

Lin et al. [8] presents a way to solve the problem of finding the most cost-effective route for a vehicle that needs to refuel. The goal is to find the best path and refueling stops to minimize the total cost of fuel. By identifying two properties of a refueling problem:

1. The path taken between any two consecutive refuel stops must be the shortest path in terms of fuel consumption (If you don’t buy fuel between two stops, a shorter route will always be cheaper) and;
2. When traveling between refueling stops, the vehicle fuel tank must be either full (U) when leaving a refueling stop or at its minimum level (L) when arriving at the next.

Khuller et al [5] considers several different refueling problems that generalize the classic shortest path and Traveling Salesman Problem by incorporating the cost of gas. The proposed model assumes a vehicle with given tank capacity, U , which is the maximum distance the vehicle can travel on a full tank. At each vertex (location), gas can be purchased at a certain price per mile.

In the paper, khuller et al. proposes an algorithm to the fixed-path gas station problem. The goal is to find an optimal set of refill stops along the path. The algorithm processes the gas stations in the order they appear on the fixed route, track the current fuel level and the cost incurred so far. Then using a priority queue (a data structure that quickly finds the minimum element) is used to store the gas stations you have passed. The stations in the queue are prioritized by their gas price. The decision to refuel is made checking this priority queue and finding the cheapest station u that you could have reached from a previous fill-up. If the fuel is not enough to reach the cheapest station u then refill just enough at current location v to reach u .

The algorithm continuously compares current station v with the stations in the priority queue. If v is cheaper than the most expensive station in the queue, it means v is a better candidate for a refueling stop. The algorithm updates the refueling plan to reflect this. This approach is very efficient and provides an exact optimal solution for a fixed-path problem. The problem becomes much more tractable than the full “Gas station problem” or the “Traveling salesman problem” with refueling constraints, which are NP-hard [5]. As a limitation, the paper stresses the underestimation of cost such as fuel consumption rates on non-highway roads that are of importance if highways are left and detours to visit cheap gas stations are accepted[5] [7].

“A generic model of Motor-Carrier Fuel Optimization” by Suzuki [9] introduces a MILP formulation proposed as a Fixed-route vehicle refueling. The proposed model is “generic” because it extends the standard fuel optimizer by setting variables for maintenance, depreciation and opportunity cost.

A variable reduction technique was proposed and extended by Suzuki [9], [10] by extending his generic MILP to account for detours:

Basic FRVRP: This is a simpler version where all fuel stations are assumed to be on the main route and there is no minimum quantity for fuel purchases.

Complex FRVRP: more realistic version that considers both “out-of-route” (which is referred to as detour distance in this report for the model), i.e. the distance a truck must divert to reach a station and a minimum refueling quantity. There are important factors for both solution quality and driver satisfaction. We focus on the latter FRVRP formulation.

Later in his work, Better bounds for inequalities in MIP constraints are proposed to tighten constraints and reduce solution space [11].

Kampker et al. in his paper for a fixed-route refueling strategy for H_2 fuel cell vehicle, focusing on the node reduction and the Mixed Integer Program (MIP) [12] by removing refuel stations outside a set beeline distance.

A state of the art review of all the Green vehicle routing problem by Asgahri et al. mentions different approaches and specific problem statements with varying objective functions, energies and approaches to solve them [4]. The author highlights the gap in the field of green transportation, most of the publications have assumed homogeneous vehicles consist of only ICEVs and EVs, which is not exactly the current picture. A different and rarely used approach is the utilization of heterogeneous fleet incorporating fewer polluting means of transport and alternative technologies.

Furthermore, few optimization approaches were developed for determining the optimal combination of different types of vehicles such as ICEVs, EVs, and HEVs (fleet size and mix problem) whose specific characteristics must be covered in adequate formulations [4].

3 METHODOLOGY

3.1.1 System Architecture

The overall system is structured as a modular pipeline, as summarized in Figure 4. Each stage corresponds to a well-defined component in the implementation.

1. Vehicle selection. The process begins by selecting a vehicle type (e.g., Autobus, Autocar, Tracteur Routier). The catalogue of vehicles, stored in the vehicle catalog table, provides attributes such as tank capacity, autonomy, and typical consumption rates (queried via `vehicle_stats.py`).

2. Route preprocessing. The customer specifies the immatriculation and the journey (`route_id`). Using OSRM, a sequence of GPS coordinates with approximately 2 km spacing is retrieved. Timestamps are then interpolated to ensure consistency, and elevation is attached from

external datasets. Cumulative distances are computed for each geolocation (nodes.py, input_builder.py).

3. Node construction and reduction. The detailed route points are transformed into nodes, enriched with attributes: load, and slope. Fuel stations are assigned to bordering nodes using beeline distance. A node-reduction strategy is applied to keep only relevant stations while preserving consumption integrity (nodes.py, kampker_nodes.py).

4. Consumption modelling. The energy consumption per edge is estimated using the consumption model, which combines physical parameters (aerodynamics, rolling resistance, slope, auxiliaries) with stochastic driving behavior. This produces a distance-weighted consumption map for the reduced node chain.

5. Optimization solver. The reduced node set, distance map, station map, and consumption map is passed to the MILP solver (solver.py). The solver implements fuel-balance constraints, driver rest regulations, and time limits. Depending on the selected fuel type (B7, B100, HVO, EV, GNV, H2), it computes the least-cost refueling strategy, accounting for detours, refuel time absorption, and contract-based energy prices.

6. Fuel station and price data. All refueling options are retrieved from the database tables fuel_stations and fuel_prices. The former stores static station attributes (location, name, power), while the latter contains year-specific price scenarios (2025–2032).

7. Results and visualization. The solver produces structured outputs through *OptimizationResult* (result_builder.py), including cost breakdown, fuel usage, time costs, detour distances, and emissions. These results are exported to Excel for further analysis and visualized on interactive maps showing the route, refuel stops, and detours. A detailed SQL table

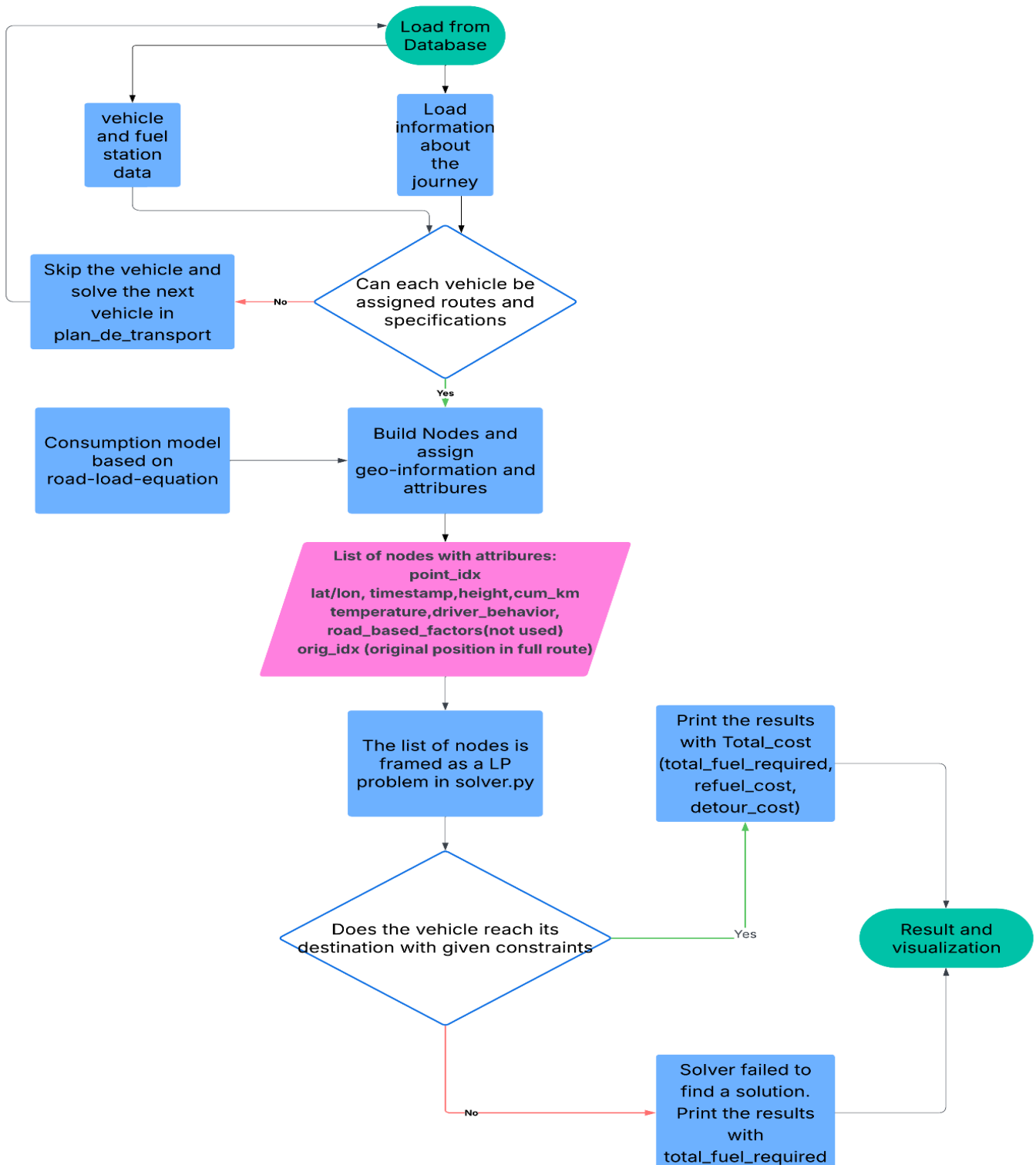


Figure 4 flowchart with overview of GreenFleet implementation

structure is described in Figure 5. Note that not all attributes and tables are explained as some tables are place holder for future work.

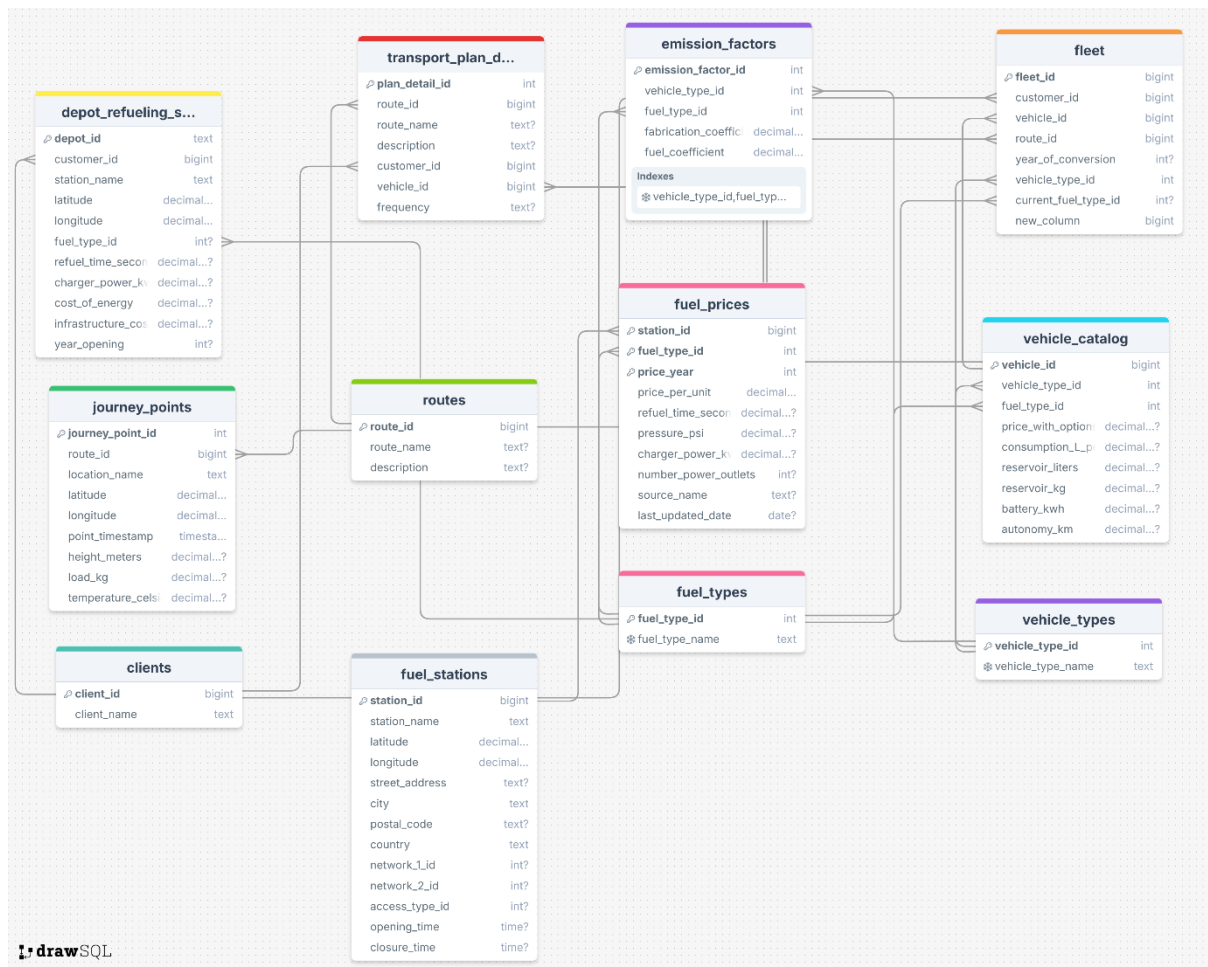


Figure 5: Database structure in PostgreSQL

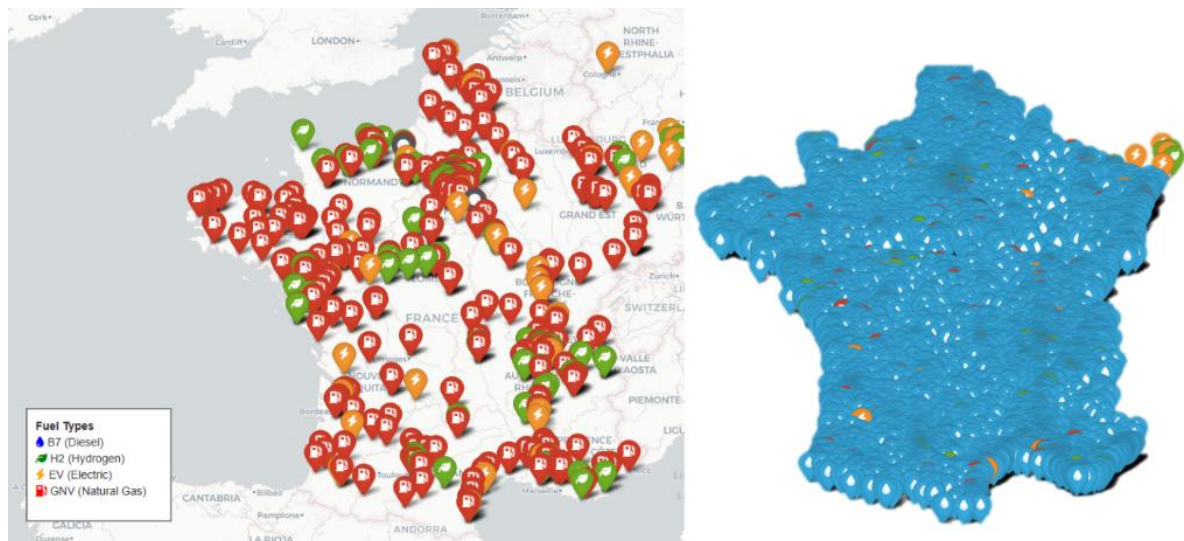


Figure 6: Mapped fuel stations for different energies across France. The image on the left has H2, EV and GNV. The image on the right includes B7

3.1.2 Input for the model

Required Inputs:

1. Vehicle type and fuel from the vehicle catalog.
2. Journey-points from the user with name, Latitude/Longitude, timestamp, elevation to compute gradient, load (in tons), road characteristics.
3. "Plan de transport" to assign each immatriculation a vehicle type from the catalog and journey points.
4. Depot refueling sites for additional home refueling sites depending on the existing and future planned sites specific to the transporter.

3.1.3 Graph Structure and node reduction (approach-2)

The route is represented as a sequence of nodes forming a path graph. Each node corresponds to a GPS trace point along the journey and is enriched with several attributes:

- Geospatial attributes: latitude, longitude, and cumulative distance (*cum_km*) from the start of the journey, obtained via the Open-Source Routing Machine (OSRM) API.
- Temporal attributes: the timestamp of the vehicle at this location.
- Environmental conditions: temperature and elevation.
- Driver and road context: driver behavior classification and scaling factors that influence energy consumption.
- Operational attributes: vehicle payload and indexing information. Edges are defined by the sequential order of nodes in the trace. An edge $(i, i + 1)$ is associated with the driving distance ($cum_km[j] - cum_km[i]$) and a segment-level fuel or energy consumption rate. These edges serve as the fundamental building blocks for the MILP solver.

In discussing the mixed-integer programming (MIP) formulation of the FRVRP, Suzuki in his work estimated that with n truck stops, the model requires n binary decision variables. This results in 2^n possible combinations of truck stops, causing the solution space to grow exponentially with n . As a consequence, even moderate values of n leads to very large problem instances, making the search for optimal solutions computationally challenging [10]. He reports that for cases where $n > 200$, obtaining optimal solutions through standard mixed-integer linear programming techniques becomes inefficient [9].

A more robust approach is to apply a node reduction algorithm is taken from the work of Kampker et al [4]. Instead of retaining all GPS points, the sequence is compressed into a reduced set of border nodes that preserve the essential structure for refueling optimization:

1. For each node, the nearest fuel station is identified using haversine (beeline) distance.
2. For each station, only the single closest node is retained, while stations beyond a maximum detour threshold are ignored.
3. The reduced nodes are enriched with station attributes such as fuel price, refuel time, charger power, and detour distance, which are later used by the MILP solver.

By removing redundant intermediate points, this reduced graph preserves both fuel consumption and operational fidelity, while drastically reducing the problem size. The result is a tractable MILP formulation that focuses only on nodes where refueling decisions can meaningfully occur.

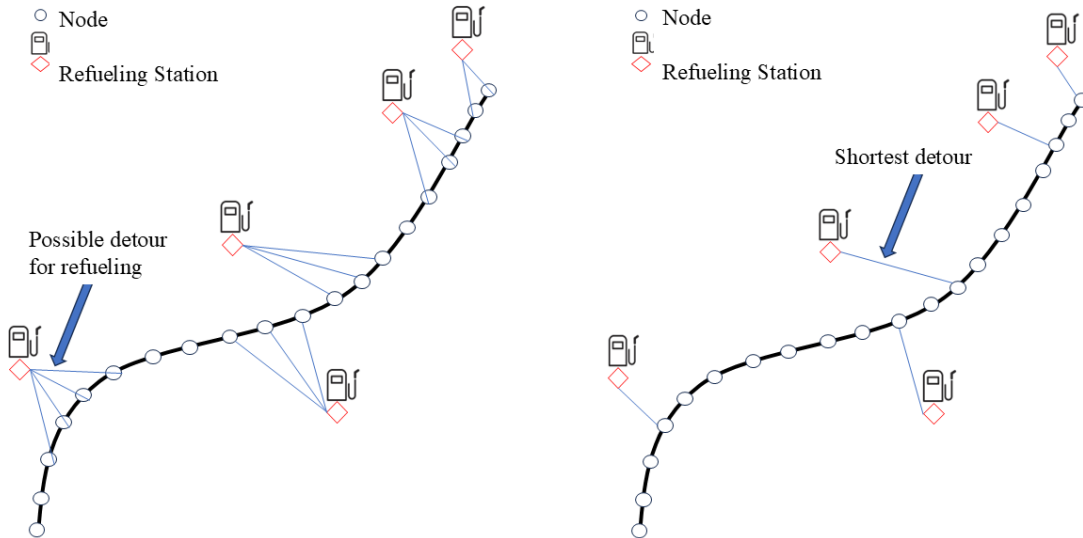


Figure 7: Node reduction as described by Kampker et al. (2023)[12]. The closest refueling stations to each node on the route (left); closest node to a refueling station among the neighborhood of selected nodes

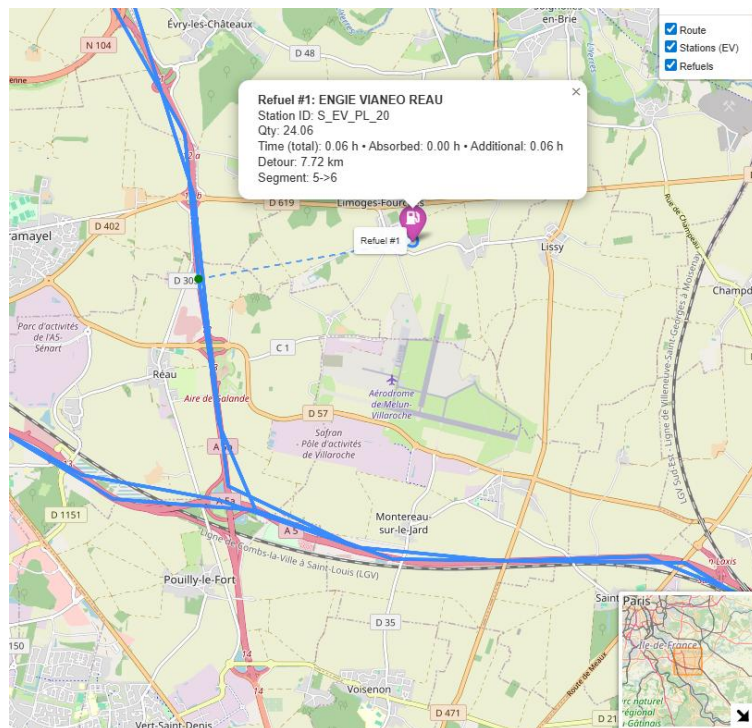


Figure 8: bq-265-be_2025-06-02_2028_EV

3.1.4 Modeling consumption of a vehicle

To include a model of energy consumption into the model, the required energy for a vehicle has to be determined. The data to compute forces come from the user defined speed, weight and gradient associated with the node (n) and additional factors such as environmental factors (wind, temperature, humidity) and other fixed parameter associated with the route like type of the road which are a separate input to the model used for consumption. The forces considered are the Inertia {Kinetic Energy} (1), the slope {Potential Energy} (2), the Aerodynamic drag (3), and the rolling resistance (4) and Auxiliaries (5) of the truck.

$$\Delta E_{\text{kin}} = \frac{1}{2} m (v_2^2 - v_1^2) \quad (1)$$

$$\Delta E_{\text{pot}} = mg(h_2 - h_1) \quad (2)$$

$$E_{\text{aero}} = \frac{1}{2} \rho C_d A_f v_{\text{air}}^2 d \quad (3)$$

$$E_{\text{roll}} = mg C_{rr} d \quad (4)$$

$$E_{\text{aux}} = P_{\text{aux}} \Delta t \quad (5)$$

Here m denotes the vehicle mass, V_1 and V_2 the initial and final velocities, g the gravitational acceleration, h_1 and h_2 the elevations at consecutive nodes, ρ the air density, C_d the drag coefficient, A_f the frontal surface area, v_{air} the relative air speed, d the traveled distance, C_{rr} the rolling resistance coefficient, P_{aux} the auxiliary power demand, and Δt the travel time for the segment. The total energy demand of the vehicle over an edge is obtained as the sum of these contributions, adjusted by drivetrain efficiency and recuperation effects (for electric or hydrogen vehicles), as described in the consumption model.

P_{aux} (auxiliary power demand) depends on factors such as radiator design, cabin HVAC and electrical components. A detailed impact of auxiliary power has not been performed and a constant value is assumed in this report. The consumption model used can be found in appendix 7.1.5

3.1.5 Estimating Emissions

Based on ADEME emission factors and an internal lookup table (vehicle type, fuel type and fuel_specific), we use two coefficients: the fabrication coefficient k_{fab} ($KgCO_{2e}/km$) and the fuel coefficient k_{fuel} ($KgCO_{2e}$ per unit of energy carrier). For each optimized route, fabrication emissions are $E_{fab} = k_{fab} \times Distance (km)$ and fuel-use emissions are $E_{fuel} = k_{fuel} \times Total\ fuel\ required$, where k_{fuel} is expressed in $kgCO_{2e}/L$, $kgCO_{2e}/kWh$, or $kgCO_{2e}/kg$ depending on the energy. Refer appendix 7.1.4 for the table

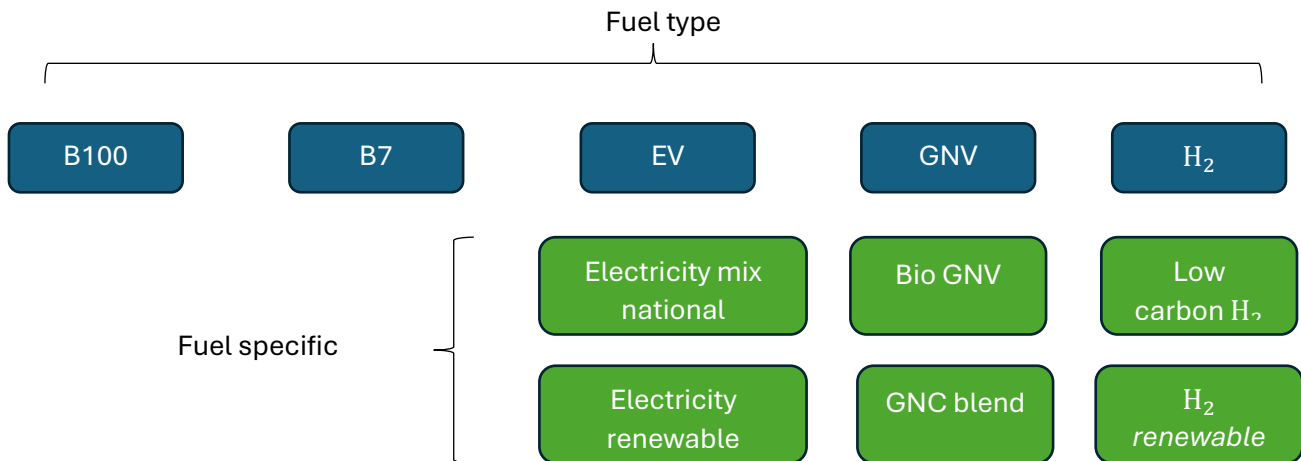


Figure 9 Each fuel type can be broken down into "fuel specific" which has different emission factors depending where this energy is sourced from.

	fuel_specific
	B100
	B7
	Bio GNV
	Electrique mix national (méthode moyenne 2020)
	Electrique renouvelable

3.1.6 Setting price of fuel and time it takes to refuel

Due to lack of precise data, the prices at each fuel station are a constant which changes over years. The refuel time parameter (time it takes to refill one unit of fuel) is constant for an energy, except for EV where refuel time depends on the kWh rating of the charger. The MILP weighs in the refuel cost and time-cost/cost to make a detour into consideration to get the least cost refuel plan.

3.2 MILP Setup and simplification

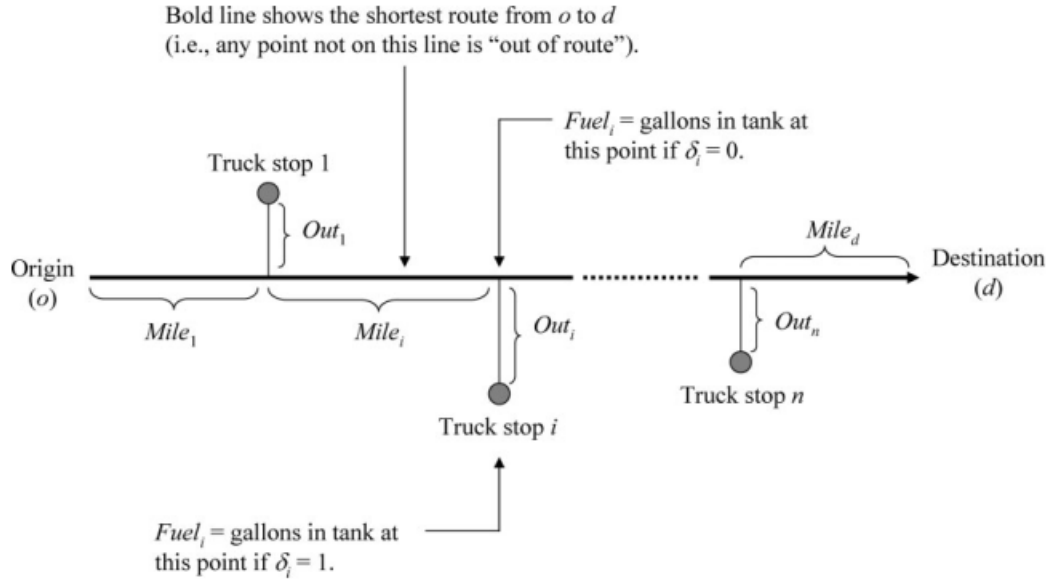


Figure 10: Schematic representation of the route, δ is the decision variable. Source (Suzuki 2008)[9]

A general MILP formulation for solving refueling problem is taken from (Suzuki 2008)[9]. The following factors are considered for setting up the MILP [9]:

1. truck's tank capacity (TC),
2. trip starting fuel (SF),
3. trip ending fuel (EF),
4. fuel consumption rate (miles per gallon) (FC),
5. minimum fuel to be maintained at all times (LF), and
6. out-of-route distance to each candidate truck stop (the extent to which a vehicle must deviate from the optimal route to reach the truck stop).

A notable feature of this model is that it uses different fuel consumption rates for "on-route" highway segments versus "out-of-route" non-highway segments. This is an improvement over commercial optimizers that use a single MPG for the entire trip) [9]. Using the binary decision δ_i and amount of fuel purchased ϕ_i as variables and the constraints:

The fuel level in the tank at any time (FL) should be greater than minimum fuel level (LF)

1. At the end of the trip, $FL \geq LF$
2. If a truck refuels at a station, a minimum amount (MP) of fuel has to be purchased
3. Refueling at any point cannot exceed tank capacity (TC)

4. Constraint on Fuel level dynamics at point $(i, i + 1)$;

$$FL_{i+1} = FL_i - FC \times Distance + \emptyset_i * \delta_i; \delta_i \in \{0,1\}$$

Few modifications were made to the MILP. The first approach was to solve the Linear problem segment-wise, this is elaborated in section 4.1. Another, more accurate approach for modified MILP with reduction technique adopted from [12] is presented in section 4.2

The MILP formulation is solved under a set of constraints that ensure physical feasibility, operational realism, and consistency of the solution. At the very beginning of the journey, the fuel level at the first node is fixed to the initial fuel level, which is provided as a user input. From this point onwards, the model enforces a strict fuel balance at every step of the route: the fuel available at the start of an edge, minus the fuel consumed while traversing it (and any additional detour fuel costs), plus the quantity refueled at a station, must exactly equal the fuel level at the end of that edge. This mass-balance formulation guarantees consistency across all nodes. To prevent unrealistic behavior, additional constraints are introduced. The fuel level at any node can never exceed the tank capacity, and the amount refueled is similarly bounded so that the vehicle cannot take on more fuel than the available headroom in the tank. For instance, if the vehicle has a 100 L tank and arrives at a node with 80 L, the maximum permissible refuel quantity is 20 L. At the other end of the spectrum, fuel variables are also given a strict lower bound, corresponding to 20% of the tank capacity, so that the model never allows the vehicle to operate below a critical reserve level.

The detour structure requires special attention. If the optimizer decides to make a detour to a fuel station, it must ensure that the vehicle has enough fuel to complete this detour. Moreover, a logical linking condition is applied: a detour is only valid if it is accompanied by refueling at that station. This avoids meaningless solutions where a detour is taken without actually purchasing fuel.

Refueling times are modeled in a technology-specific manner. For conventional fuels, the time required is calculated as the refuel quantity multiplied by a station-specific parameter `timePerUnitFuel` (e.g., hours per liter or hours per kilogram). For electric vehicles, refueling corresponds to charging, and each station has an associated power rating that directly determines charging time. In both cases, the time required to refuel is explicitly linked to the decision variables.

Continuity is also preserved across the journey: the fuel level at the end of one edge becomes the starting fuel level of the next, thereby ensuring seamless propagation of energy states along the route. Finally, global constraints tighten the overall solution. At the end of the journey, a minimum fuel level must be respected to avoid unrealistic empty arrivals. In addition, the total trip duration — which aggregates driving time, detour time, and any additional refueling delays not absorbed by breaks — must remain within a predefined upper bound. If this total time exceeds the specified limit, the solution is deemed infeasible. Together, these constraints ensure that the MILP captures both the operational requirements of transport and the physical limitations of the vehicle.

A simplification to break down the larger route into smaller MIP problems, we break them into smaller journeys. This makes solution space much smaller as we solve for fewer points. To ensure we don't lose information, they all the journeys are aggregated again to the route equivalent.

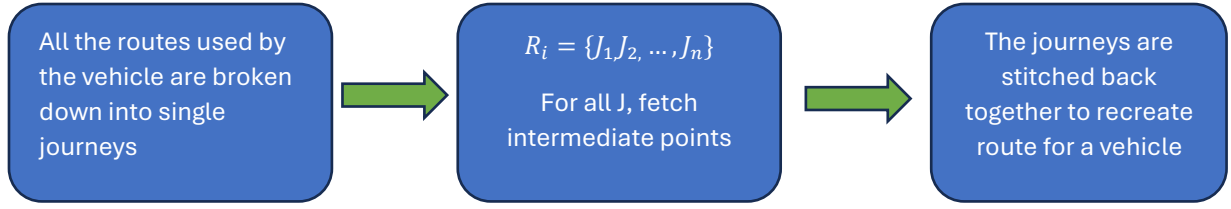


Figure 11 Bigger route is broken down into smaller journeys. These journeys are solved in the MILP. The results are then compiled again to represent a route

Keeping the MILP formulation identical with same objective but slightly modified constraints, a model was developed to solve the MILP problem using station projections on the segment.

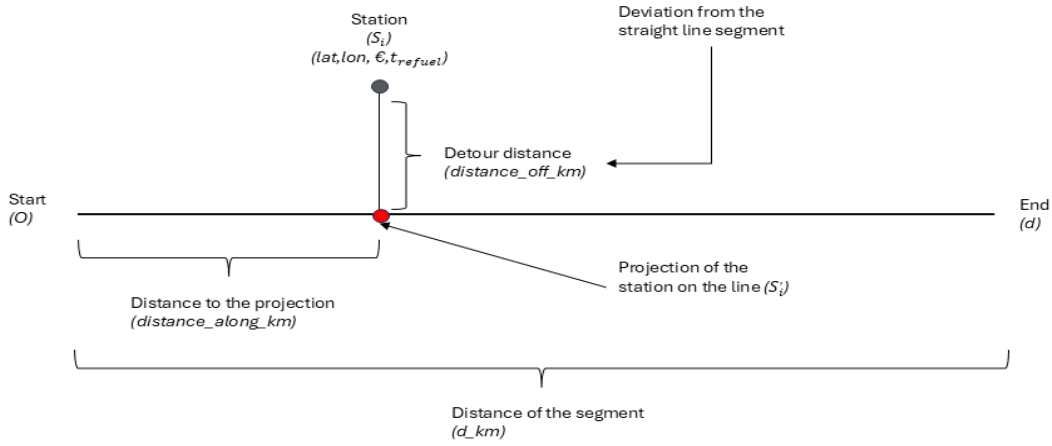


Figure 12: A segment of a route. A set of segments in order of time represent a journey. $start(o)$ and $End(d)$ are the points of a segment.

For a station S_i , its perpendicular projection was dropped on the segment. This point, S'_i is $distance_along_km$ away from the start of the segment, total distance of which is d_km . The distance from S'_i to S_i is marked as $distance_off_km$. This is the deviation from the straight line segment.

A journey is broken down into smaller straight line segments. We set a MILP model to solve the set of these segments. Here is the description of the model:

Sets and Indices

- $N = \{0, 1, \dots, n - 1\}$ nodes ordered along the fixed route.
- Segments $\mathcal{E} = \{(i, i + 1) : i \in \{0, \dots, n - 2\}\}$
- For each segment $i \rightarrow i + 1$, candidate stations S_i

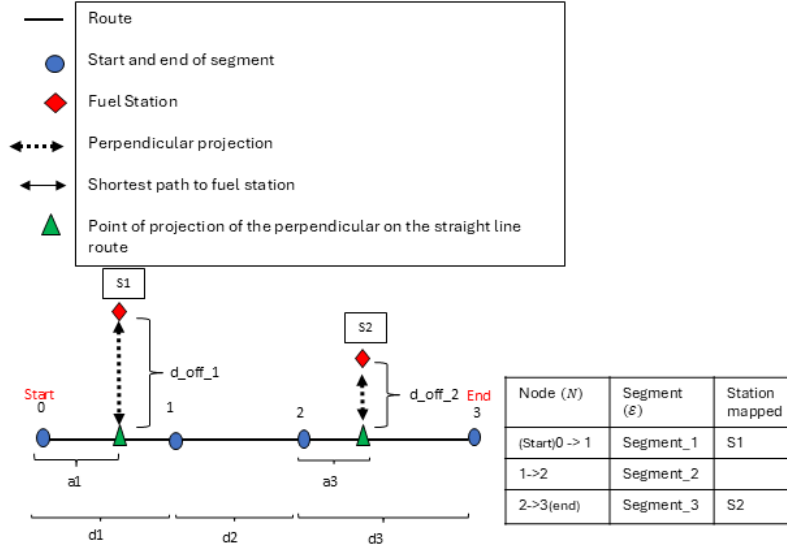


Figure 13: Visualizing sets of nodes, segments and stations on a route

Parameters

1. d_i (km) : route distance on segment $i \rightarrow i + 1$,
2. C_i (L/km, kg/km, kwh/km): consmption rate of fuel on segment $i \rightarrow i + 1$,
3. C_{drv} (€/hr): Driver cost per hour
4. v_i (km/hr) : average speed on segment $i \rightarrow i + 1$,
5. $v^{\det}$ (km/hr) : average detour speed when a detour is made $S_i \leftrightarrow S'_i$,
6. B_i (hr) : available break time on segment $i \rightarrow i + 1$,
7. $P_{i,s}$ (€/L, €/kg, €/kWh) : unit fuel/energy price at station $S \in S_i$,
8. a_i (km) : distance along the route from start point of the segment to the projection of the station on the route $i \rightarrow S'_i$,
9. $b_{i,s}$ (km) : distance off the route (one-way) to station S
10. γ_i (L/km, kg/km, kwh/km): detour consumption rate for vehicle when performing a detour,
11. F (L for B7, B100 ; kg for GNV, H_2 ; kWh for EV) : Tank or battery capacity for a vehicle
12. F_0, F_{min}^{end} (L for B7, B100 ; Kg for GNV, H_2 ; kWh for EV) : initial and final fuel level
13. T_{max} (hr) : total driving time limit
14. $T_{i,s}$ (L/hr for B7, B100 ; kg/hr for GNV, H_2) : refuel time at a station, for EV the time to recharge can be computed from the kWh rating of the charger
 - a. EV: charger power $P_{i,s}$ (kW); refuel time = $\frac{q}{P_{i,s}}$
 - b. Non-EV: time per unit $\tau_{i,s}$ (h per fuel unit); refuel time = $\tau_{i,s} q$
15. M^{ref} : tight big-M for refuel time linking

Decision Variables

1. f_i^{before} (L, kg, kWh) : fuel before segment $i \rightarrow i + 1$; bounds $[0.2F, F]$

2. $f_i^{after}(L, kg, kWh)$: fuel after segment $i \rightarrow i + 1$; bounds $[0.2F, F]$
3. $t_i^{travel}(hrs)$: travel tiome on route segment $i \rightarrow i + 1$

For each $S \in S_i$:

- $x_{i,S} \in \{0,1\} (km)$: route distance on segment $i \rightarrow i + 1$
- $q_{i,S} \geq 0 (L, kg, kWh)$: quantity refueled at station S
- $t_{i,S}^{ref}(hrs)$: total refuel time (time it takes to refuel)
- $t_{i,S}^{abs} \geq 0 (hrs)$: refuel time absorbed by break
- $t_{i,S}^{add} \geq 0 (hrs)$: additional refuel time; this is the time which could not be adjusted with break.
- $t_{i,S}^{det} \geq 0 (hrs)$: detour time

Initialization

$$f_0^{before} = F_0. \quad (1)$$

Segment Physics & Linking

$$t_i^{travel} = \begin{cases} \frac{d_i}{v_i}, & v_i > 0, \\ 0, & v_i = 0 \end{cases} \quad (2)$$

$$f_i^{after} = f_i^{before} - C_i d_i - \sum_{S \in S_i} (2b_{i,S} \gamma x_{i,S}) + \sum_{S \in S_i} q_{i,S} \quad (3)$$

Tank capacity after segment:

$$f_i^{after} \leq F. \quad (4)$$

Link to next segment:

$$f_i^{after} = f_{i+1}^{before} \quad (5)$$

Detour Feasibility & Refuel Headroom

$$f_i^{before} - C_i a_{i,S} \geq 2 b_{i,S} \gamma x_{i,S}, \quad \forall i, \forall S \in S_i \quad (6)$$

Headroom

$$q_{i,S} \leq (F - (f_i^{before} - c_i a_{i,S})), \quad \forall i, S. \quad (7)$$

Active refueling only if detour

$$q_{i,S} \leq F x_{i,S}, \quad \forall i, S. \quad (8)$$

Refuel Time, Detour Time, and Decomposition

$$t_{i,S}^{ref} = \begin{cases} \frac{q_{i,S}}{P_{i,S}}, & EV \text{ (charger power known)} \\ \tau_{i,S} q_{i,S}, & non - EV \end{cases}, \quad \forall i, S \quad (9)$$

$$t_{i,S}^{det} = \frac{2 b_{i,S}}{v^{det}} x_{i,S}, \quad d_{i,S}^{det} = 2 b_{i,S} x_{i,S}, \quad \forall i, S. \quad (10)$$

$$t_{i,S}^{abs} \leq B_i + M^{ref} (1 - x_{i,S}), \quad (11)$$

$$t_{i,S}^{abs} \leq t_{i,S}^{ref}, \quad (12)$$

$$t_{i,S}^{add} \geq t_{i,S}^{ref} - B_i - M^{ref} (1 - x_{i,S}), \quad (13)$$

$$t_{i,S}^{add} \leq M^{ref} x_{i,S}, \quad (14)$$

$$t_{i,s}^{add} \geq 0, \quad (15)$$

$$t_{i,s}^{ref} = t_{i,s}^{abs} + t_{i,s}^{add}, \quad \forall i, S. \quad (16)$$

End-Fuel Requirement

$$f_{n-1}^{before} \geq F_{min}^{end}. \quad (17)$$

f_{n-1}^{after} is the fuel level at the end

Total Time and Limit

$$T_{tot} = \sum_{i=0}^{n-2} t_i^{travel} + \sum_{i=0}^{n-2} \sum_{s \in S_i} t_{i,s}^{det} + \sum_{i=0}^{n-2} \sum_{s \in S_i} t_{i,s}^{add} \quad (18)$$

$$T_{tot} \leq T_{max}. \quad (19)$$

Objective

$$\min. \underbrace{\sum_{i=0}^{n-2} \sum_{s \in S_i} p_{i,s} q_{i,s}}_{Refuel\ cost} + \underbrace{C_{drv} T_{tot}}_{Time\ cost} \quad (20)$$

Bounds & Domains

$$0.2F \leq f_i^{before} \leq F, \quad \forall i. \quad (21)$$

$$0.2F \leq f_i^{after} \leq F, \quad \forall i \leq n-2. \quad (22)$$

$$q_{i,s} \geq 0, t_{i,s}^{ref} \geq 0, t_{i,s}^{abs} \geq 0, t_{i,s}^{add} \geq 0, t_{i,s}^{det} \geq 0, d_{i,s}^{det} \geq 0, \quad \forall i, S. \quad (23)$$

$$x_{i,s} \in \{0,1\}, \quad \forall i, S. \quad (24)$$

Objective (20) minimizes the refueling and time costs. Constraints (21),(22) ensure that atleast 20% of the tank capacity/battery charge F is at each point of the route. Constraint (6),(7),(8) check the feasibility of detour (fuel available at projection point must cover the round-trip detour fuel if we chose to detour), meaning we should have enough fuel to reach a station if a detour were to be made. Refueling is allowed only if a detour is taken ($x_{i,s} = 1$). The quantity refueled is bounded by available headroom. There is a constraint on time (18)(19) where the sum of all the duration should not exceed Total driving time limit T_{max} . Constraint (1) and (17) set the start and minimum level of fuel at the end of a route. Fuel balance is preserved with constraint (3) where fuel at the end of a segment should be equal to the fuel at the start of the segment, any quantity refueled and consumption to reach from start of the segment to the end. Then the fuel level at the end of a segment at node (i) is assigned as the starting fuel level at the same node but to the adjacent segment in constraint (5). A simplified example is given in Figure 14

Node (N)	Segment ($\mathcal{E}$)	Fuel before	Fuel after	Fuel consumed
0	-	100	100	-
1	Segment_1	80	120	20
2	Segment_2	120	-	20

Table 1: Example fuel-balance summary for given route

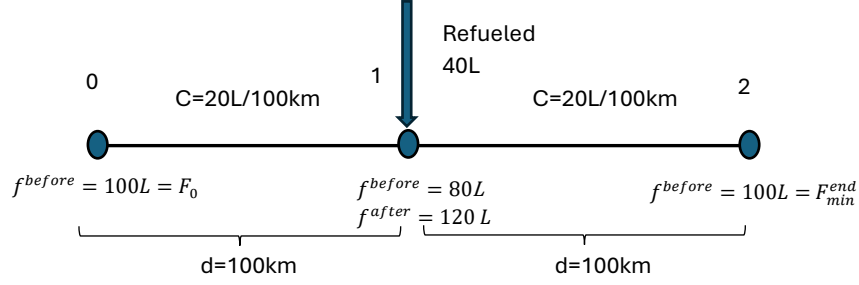


Figure 14: Example of fuel balance. A consumption of 20L/km is assumed over a route broken into 2 segments of 100km each. The arrow at node-1 indicate 40L of fuel was added. The detailed balance is shown in the table above

Constraint (6) checks whether we have enough fuel to at the start of a segment to reach the projection and make a detour with constraint (7) ensuring we only refuel the available space in the tank/battery.

In this formulation in constraints (9,10,11,12,13,14,15,16), only travel time, detour time, and additional refueling time enter the MILP directly. The total time is accounted “post-solve”, since the exact driving schedule is not known we assume that it is preferred if the refueling happens during break. A driver has to take break as per (EC) No 561/2006 [13]. A simple implementation of a 40min break after 4 hrs. of driving is simulated. The time is added post solve and we do not double count time if refueling and break overlap.

3.2.2 Limitations

Due to mapping of stations onto segments, the size of the problem increases significantly. Even if the number of stations are reduced. The straight line approximation of a route creates distortion, the straight line gives unrealistic approximation of routes and the detour criteria is weak. It can foresee fuel sufficiency and the the detour decision is made only at the start of a segment. Few mapped routes have been presented in the appendix 7.1.4 .An improved model is proposed in following section.

3.2.3 MILP model based on nodes and distances (Approach 2)

We with change in data a node carries, the MILP model can be improved. Instead of mapping stations to a segment, we can assign them to the nodes. The condition being we have a good sample of nodes at a uniform distance and a sample coordinate at every ~2kms or less. The set, parameters, constraints and objective remain the same but we don't solve for all the edges (segments in previous case)

stations are “exposed” on edge $(i, i + 1)$ and refueling is modeled as happening after finishing that edge (at node $i + 1$). The parameter a_i (km) (distance along the route from start point of the

segment to the projection of the station on the route) is not required. Instead for a route, we calculate cumulative distance cum_km from the starting to the end node.

We reduce the decisions by completely eliminating where refuel was not intended anyways.

In the segment model, every small segment with its fuel balance equation and decision variable had to be solved

In node based model, we compute consumption for every edge between nodes $(i, i + 1)$ that enters the balance: $f_i^{after} = f_i^{before} - C_i d_i - \sum_{S \in S_i} (2b_{i,s} \gamma x_{i,s}) + \sum_{S \in S_i} q_{i,s}$. So even if no station is nearby, consumption is still accounted.

Nodes without stations

- If a edge between nodes $(i, i + 1)$ has no feasible stations, then there are no decision variables for it. The only thing that happens there is consumption subtraction:

$$f_i^{after} = f_i^{before} - C_i d_i$$

so the balance just “carries forward” fuel.

All edges contribute consumption (fuel decreases continuously). Only edges with eligible stations give the solver the choice to refuel/detour. The MILP only has to “decide” at those points; the rest are passive consumption updates.

Criteria for detour feasibility is modified

Before the optimization problem is solved, the project reduces the number of candidate stations using a border-node reduction strategy implemented

- Each route is discretized into a sequence of nodes with cumulative distance (cum_km).
- For each station, the algorithm finds its **closest route node** using the haversine distance.
- The closest node per station “area” is retained, while redundant nodes are removed.
- Each station is projected into a **PumpOnRoute** object, carrying:
 - cum_km : location along the route,
 - off_km : radial detour distance,
 - price, time_per_unit_fuel, charger_power_kw.

The projected detour distance is approximated as:

$$detour = 2 b_{i,s} \delta$$

here δ is the **safety factor** (default 1.31) applied to beeline distance. These pumps are exposed to the solver via the station map:

$$station_map[(i, j)] \rightarrow \{PumpOnRoute1, \dots\}$$

4 RESULTS

The developed model to estimate operating cost based on consumption was first tested with a client data for 138 Tracteur Routier and Porteur fleet.

Five vehicles with following immatriculation were chosen to demonstrate the working of the model:

Immatriculation	Vehicle Type	Starting fuel level	minimum fuel level at the end of journey
bq-265-be	Tracteur Routier	75% of tank capacity	60% of tank capacity
bv-803-fd	Tracteur Routier	75% of tank capacity	60% of tank capacity
cd-768-yp	Tracteur Routier	75% of tank capacity	60% of tank capacity
cf-417-wa	Tracteur Routier	75% of tank capacity	60% of tank capacity
cq-596-pm	Tracteur Routier	75% of tank capacity	60% of tank capacity

Vehicle characteristics are as follows for type “Tracteur Routier”, applicable to all the vehicles simulated:

Driver cost per hour: € 28/*hr*

Maximum detour distance: 100 *kms* (back and forth)

Tank capacity is as follows: 800*L* for B7, B100. 250.4 *kg* for GNV, 542.8 *kWh* for EV , 43.8 *kg* for *H*₂

Based on this information, intermediate points were fetched to construct nodes along a route with each node at uniform spacing of ~2*kms*. Creating nodes with uniform space is important, if a route is generated with uneven spacing, the segment of the route with less density of nodes would not have the shortest distance, impacting the eligibility of a vehicle for a given energy.

4.1.1 Checking Eligibility of an energy for a route

The table below summarizes which vehicle was unable to complete its journey from origin to destination and for which energy. Out of all the Fuel Type simulated, Tracteur Routier on an EV platform is unable to complete certain journeys. This means there is an uncertainty whether an EV powered Tracteur is suitable for this journey, making then not eligible for consideration (the solver was unable to find an optimal solution).

The reason for infeasibility could be:

- Maximum detour distance is not suitable: Unlike B7, other sources of fuel are not widespread throughout France. If a vehicle makes a journey with no nearby fuel stations for alternate energies or it can identify the fuel stations but the constraint on detour is violated then the vehicle-energy-route combination could be ineligible.
- Operational time exceeded: Since EV, H2 and GNV can be either fast charging or slow has an impact on the operation of the vehicle for a fleet owner. It could be possible that the route only has slow charging stations available and we exceed the operational time set as a constraint.
- Not setting up Depots for refueling: Fleet operators sometimes have a dedicated depots available for refueling. Heavy vehicles using B7 as a fuel rarely use out of the network fuel stations or they undertake refueling activity themselves by using their own depots.

We made a strong assumption that our journeys throughout a month starts and end at a depot unless there is a destination close by and return to depot is much higher than staying on the road and continuing the journey on to the destination on the next day. Depots play a huge role as they are considered as a fuel station on the node. This means they are always available, the ineligible *route id* presented in the table XXX below fail mostly due to the minimum fuel level at the end of a journey is not being met, this can be resolved by assigning certain nodes as depots which offer refueling.

Status	Immatriculation	Route ID	vehicle type	Year	Fuel Type
Infeasible	bv-803-fd	bv-803-fd_2025-06-23	Tracteur Routier	2025	EV
Infeasible	bv-803-fd	bv-803-fd_2025-06-24	Tracteur Routier	2025	EV
Infeasible	bv-803-fd	bv-803-fd_2025-06-30	Tracteur Routier	2025	EV
Infeasible	cd-768-yp	cd-768-yp_2025-06-11	Tracteur Routier	2025	EV
Infeasible	cd-768-yp	cd-768-yp_2025-06-18	Tracteur Routier	2025	EV
Infeasible	cd-768-yp	cd-768-yp_2025-06-19	Tracteur Routier	2025	EV
Infeasible	cd-768-yp	cd-768-yp_2025-06-20	Tracteur Routier	2025	EV
Infeasible	cd-768-yp	cd-768-yp_2025-06-24	Tracteur Routier	2025	EV
Infeasible	cf-417-wa	cf-417-wa_2025-06-04	Tracteur Routier	2025	EV
Infeasible	cf-417-wa	cf-417-wa_2025-06-06	Tracteur Routier	2025	EV
Infeasible	cq-596-pm	cq-596-pm_2025-06-12	Tracteur Routier	2025	EV
Infeasible	cq-596-pm	cq-596-pm_2025-06-13	Tracteur Routier	2025	EV

Table 2 vehicles and the journey for which the solver could not find a solution (status = infeasible)

Currently, these vehicles use B7 as their energy. The condition to check whether a vehicle type with given attributes and constraint is able to make a journey from origin to destination for a given route is its eligibility of the vehicle.

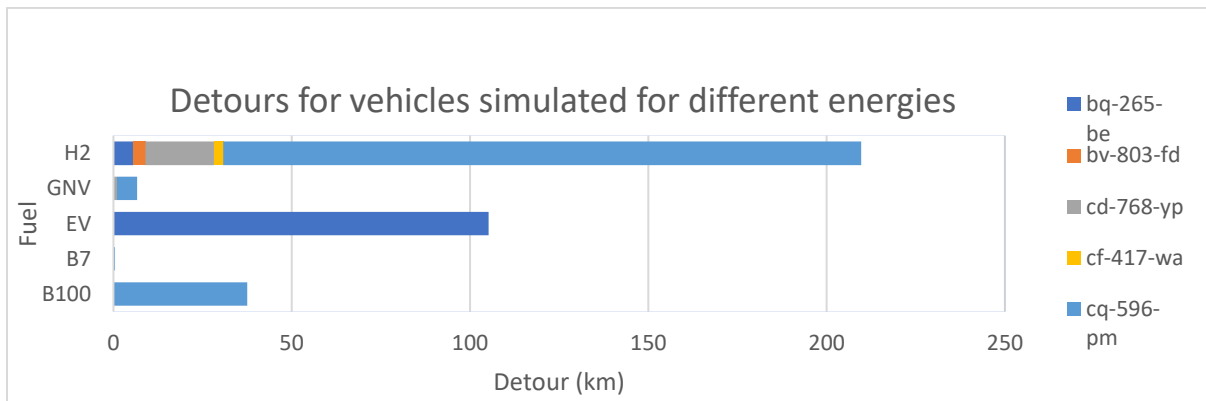


Figure 15: Detour for vehicle simulated for H2,GNV,EV,B7 and B100 from 2025-2032.

In Figure 15, the five vehicles are simulated for various energies and the H2 has the highest detour (deviation) when it starts from origin and has to go out of its way to refuel. The detour distance depends on the journey we want to simulate. Cities and centers of population have higher density of refueling stations than in the middle of nowhere but France has only 80 H2 refueling stations[14] compared to over 9000+ B7 stations, record of only 46 high power EV stations (P>150kW) , and 405 GNV stations. B100 is assumed to be only available for

commercial consumption for fleets in their assigned depots. Marolles, Oiselle and Saran are 3 locations which are known to be depots as per the client and hence the detour happens for B100 when the vehicle needs to come to the depot during a journey and limits the operability of B100 tracteur Routier.

4.1.2 Operational Expenditure for refueling activity

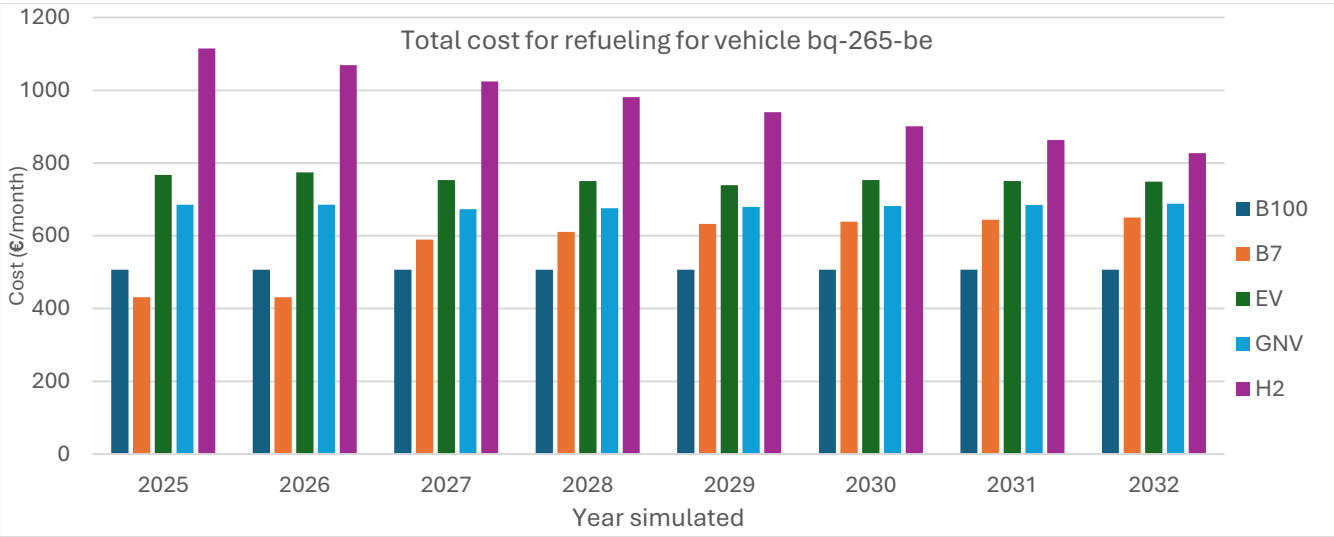


Figure 16 Total cost of refueling for vehicle bq-265-be

The Figure follows the general trend of H_2 getting cheaper and B7 and GNV getting slightly more expensive every year. The price of Electricity is assumed to be constant as French national mix is fairly consistent, meaning no change in price is expected. B100 (Bio fuel) is assumed to be only available to fleet owners and so the price is kept constant.

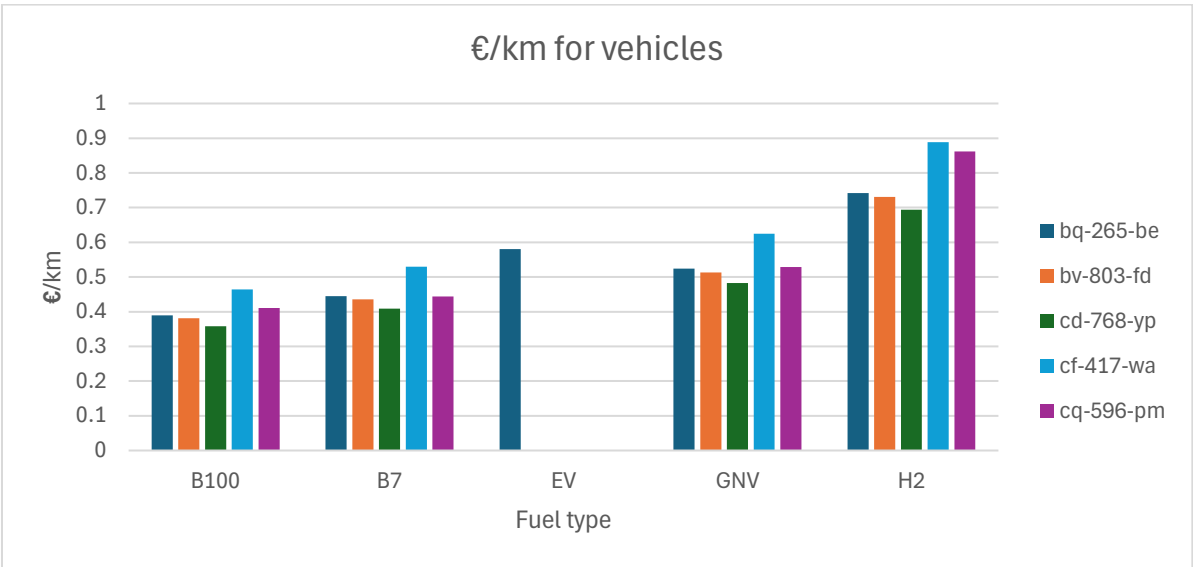


Figure 17 cost per km for all the vehicles. The energy is said to be eligible if all the journeys in that route are solved. In this case, EV is not eligible for few vehicle-route combination

If we were to find the average cost of refueling / total distance over the years 2025-2032, we see observe an interesting result. Even if the cost of electricity is low, it takes an additional time to charge an EV vehicle.

The lack of EV stations suitable for heavy vehicles means it has to make a significant detour from the route to recharge and there is a consumption cost involved when making a detour. To be more cost competitive, more EV stations have to be made available for heavy vehicles.

Even if the price to recharge is less, the associated costs balloon up which make EV unattractive as of now. Refer Table 3 for detailed breakdown

Even with a 150kWh charger, the average battery size considered for a tracteur Routier is ~550kwh. It takes ~3.5 hrs. of time to charge an EV vehicle and with € 28/hr. time cost, refueling time becomes a significant concern. This can be addressed by adding EV charging facilities at the depot where the vehicle is parked when not in used.

Year	Fuel Type	Total Cost (€)	Refuel Cost (€)	Refuel_Qty	Refuel Time (h)	Detour (km)	Total fuel required	Refuel Time Cost (€)	Detour Time Cost (€)	Fuel Detour Cost (€)	cost/km (€/km)
2025	EV	177.0629	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.152078
2026	EV	177.8531	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.15722
2027	EV	175.4974	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.141892
2028	EV	175.2291	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.140146
2029	EV	173.917	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.131609
2030	EV	175.4974	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.141892
2031	EV	175.2291	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.140146
2032	EV	174.9756	62.40218	138.6715	0.346679	73.19103	149.0954	9.707006	68.311631	31.951346	1.138497

Table 3 Cost breakdown for bq-265-be

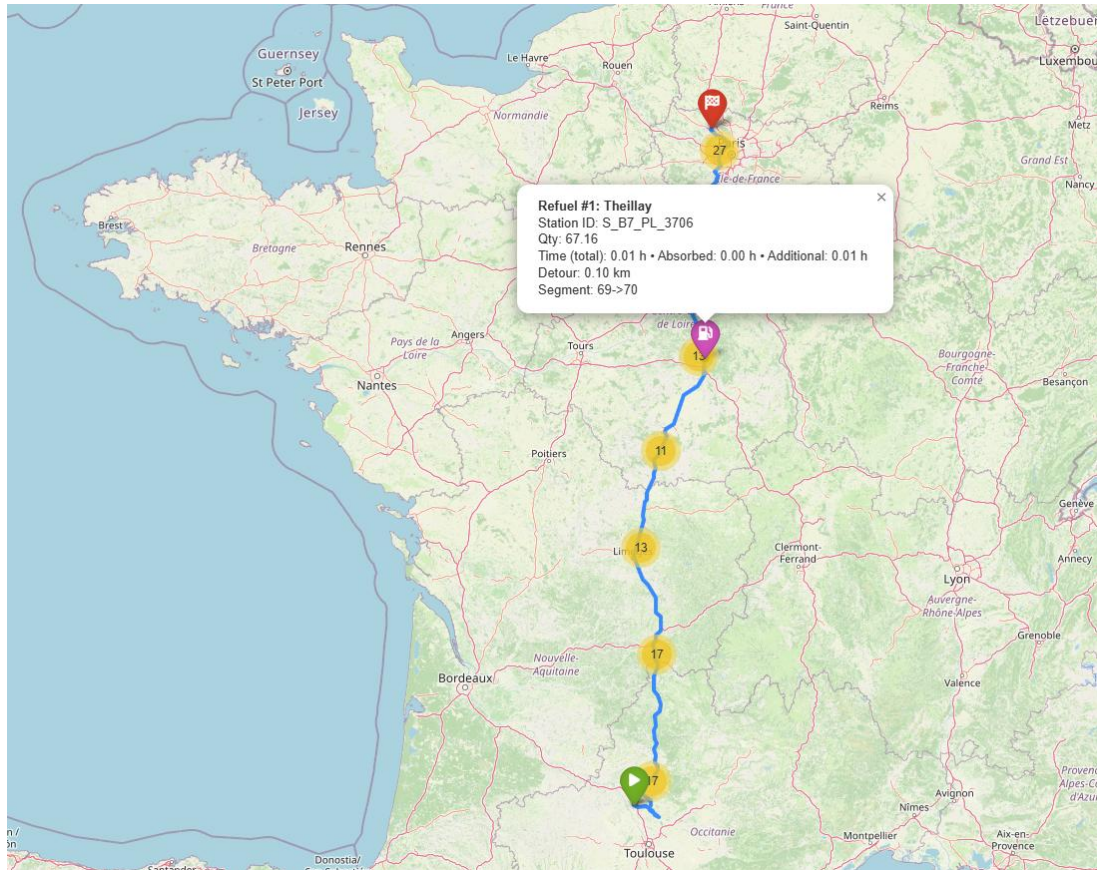


Figure 18: Example of a journey (cq-596-pm) with feasible solution.

4.2 Validation by varying cost and time parameters:

It is difficult to perform a sensitivity analysis since the fuel consumption is not polynomial to speed, speed changes for the routes and the timestamps are not uniform. The dependence of inputs makes it hard to certainly say increasing cost or time will increase the cost proportionally.

Validation through cost-detour trade:

The model correctly balances refueling costs against indirect costs of detours (time and additional fuel use). Table 4 compares two scenarios for the same EV route. In the base case, all stations are priced at €0.45/kWh, and the solver selects station S_EV_PL_20 with a short detour (11.2 km). When the price at S_EV_PL_20 is increased to €5/kWh, the solver switches to station S_EV_PL_37, despite its longer detour (20.7 km). The total cost rises from €95.7 to €112.9. This validates that the solver integrates both fuel price and detour costs in its decision-making.

Sensitivity to refuel speed:

We also test sensitivity to refueling time, relevant for EV, GNV, and H₂. Table Y shows a GNV route where all stations have the same price. In the base case, the solver chooses S_GNV_PL_235 (short detour, but slower refuel). When S_GNV_PL_350's refuel rate is doubled, the solver switches to this station: despite a slightly longer detour (0.49 km vs. 0.26 km), the faster refueling reduces driver time cost enough to lower total cost.

These simulations demonstrate that the solver is sensitive to both economic factors (price) and operational parameters (refuel speed), and makes decisions consistent with real-world expectations.

4.2.1 Validation through cost-detour trade

The optimization model captures the delicate balance between **refuel costs** and the **indirect costs of detours** (time and additional energy use). This is demonstrated through two scenarios applied to the same route:

Route ID	Base Case : S_EV_PL_20 @ 0.45 €/kWh	Price Shock : S_EV_PL_20 @ 5 €/kWh
vehicle type	Tracteur Routier	Tracteur Routier
Year	2025	2025
Total Cost	95.744	112.8689
Refuel Cost	69.8534	73.70145
Distance (km)	250.61	250.61
Refuel_Qty	155.2298	163.781
Refuel Time (h)	0.388074	0.409453
Detour (km)	11.21193	20.67247
total_fuel_required	226.5224	226.5224
Refuel Time Cost (€)	10.86608	11.46467
Detour Time Cost (€)	10.46447	19.29431
Fuel Detour Cost (€)	4.560428	8.408484

Table 4 For vehicle bq-265-be, the price of the optimal selected refuel station is changed from 0.45€/kwh to 5€/kwh

Scenario 1: The base case where each EV station has the same price 0.45€/kWh

Scenario 2: The column on the right represents a scenario where S_EV_PL_20 has been changed to 5€/kWh

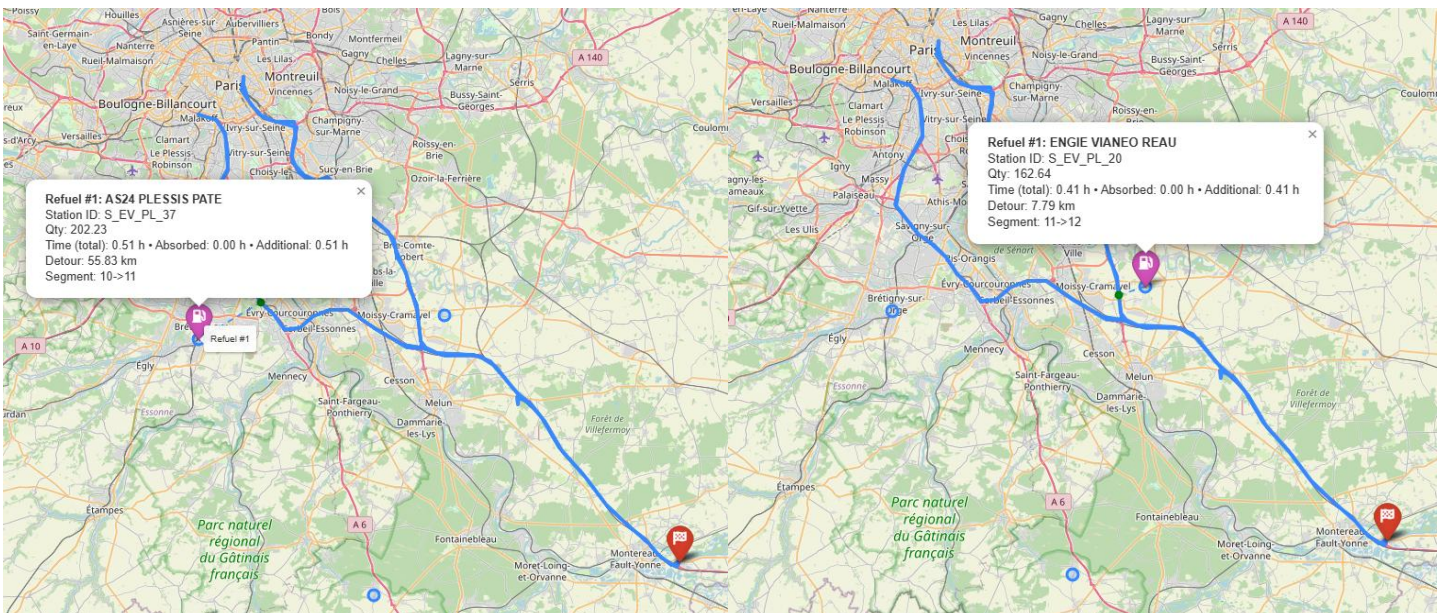


Figure 19 Base case (L) and Price shock (R).

The solver correctly identifies the trade-offs between **refuel cost**, **detour time**, and **detour fuel cost**

- **Base Case (S_EV_PL_20 selected):** The solver chooses S_EV_PL_20 because its shorter detour distance (11.21 km) and likely lower base price result in the lowest overall cost (95.74€).
- **Price Shock Case (S_EV_PL_37 selected):** When the price at S_EV_PL_20 is raised, the solver's objective function correctly evaluates the new total cost. The cost of refueling at S_EV_PL_20 (which is the refuel quantity multiplied by the new high price) becomes prohibitively expensive. The solver then finds that the next best option is to incur a longer detour (20.67 km) and refuel more energy (163.78 kWh) at S_EV_PL_37. This leads to a new, higher total cost of 112.87€.

This example highlights the model's sensitivity to **station-specific price variations**. It demonstrates that the solver does not minimize fuel price in isolation but considers the **combined trade-offs**: refuel cost, detour distance, refuel time, and driver time cost.

4.2.2 Sensitivity to refuel time

The only energies which have significant refuel time is H2, GNV and EV. It is difficult to quantify how much time it takes to refuel using GNV, we can take a guess value to check its impact on decision-making and cost.

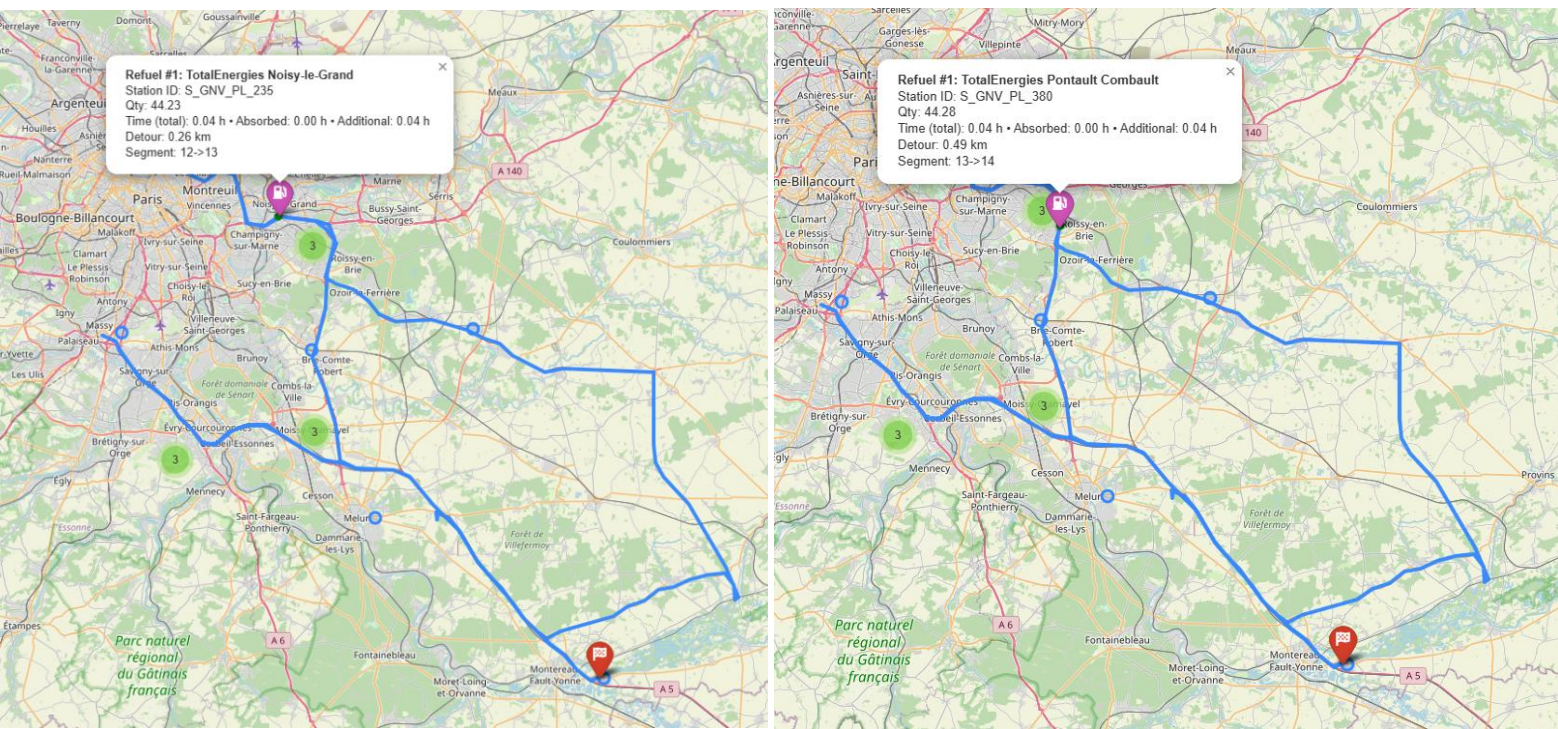


Figure 20 Base case (L) and Fast fill (R) for bq-265-be_2025-06-03 in year 2025. The choice of refuel station changes with change in refuel time

All GNV stations have same “refuel time” (quantity of fuel delivered in an hour) is set to 600kg/hr. For this route, the optimal (cheapest) station with least detour is S_GNV_PL_235. However, if the refuel_time for a station is changed, keeping every other parameter the same. The decision to refuel changes to S_GNV_PL_350.

Scenario	Base case: All stations have same rate of refueling	Fast fill GNV: S_GNV_PL_350 fills twice as fast
Vehicle type	Tracteur Routier	Tracteur Routier
Year	2025	2026
Total Cost (€/day)	92.76928816	92.06473
Refuel Cost (€)	90.22768	92.1074
Distance (km)	351.29	351.29
Refuel Qty (kg)	44.282404	44.2824
Refuel Time (h)	0.073715	0.036902
Detour (km)	0.262195	0.490652
Total fuel required (kg)	81.725948	81.72595
Refuel Time Cost (€)	2.064032	1.033256
Detour Time Cost (€)	0.244715	0.457942
Fuel Detour Cost (€)	0.124437	0.237428

Table 5 change in total cost when refuel time is doubled

At €28/hr charge incurred every time the vehicle is on the road, there is an incentive to refuel faster.

- **Base Case:** The solver selects the station with the least detour (S_GNV_PL_235). The Refuel Time Cost is €2.06. The total cost is €92.77.
- **Fast Fill GNV Case:** The Refuel Time for S_GNV_PL_350 is reduced by half. This reduces the Refuel Time Cost to €1.03. Despite the longer detour (0.491 km vs. 0.262 km), the significant savings in time cost make S_GNV_PL_350 the new optimal choice, leading to a lower **Total Cost of €92.06**.

The solver's decision to switch stations based on a change in refuel speed demonstrates that it is balancing the trade-off between **detour costs** (time and fuel) and **refueling time costs**. The savings from a faster refuel more than offset the increased cost of a longer detour, proving the model's sensitivity to this parameter.

Setting driver cost to zero will make the model independent of refuel_time. However, the detour made to reach a fuel station and the fuel we utilize to reach it will still be a critical factor, especially for H2 and EV because large detours have to be made to reach them consistently.

4.2.3 Dependence on initial and final fuel level condition

The way the MILP is framed, depending on the start and end level of fuel set as an input. The solution changes significantly. This has been highlighted in other research papers as well and we need to know or decide the start and end level based on the infrastructure around and refueling policy of the fleet we are simulating[7].

4.2.4 Dependence on Tank capacity

The reason for failure for EV to be eligible is because of limited capacity and longer refuel times. This has been a prevalent problem addressed in the literature [4]. This does not imply EVs are not feasible but because the constraints for all the energies are set the same, the difference in operation and technology is neglected.

5 LIMITATIONS AND RECOMMENDATION

5.1 Limitations

Data and Modeling Limitations

The model's effectiveness is constrained by several data and methodological limitations:

- **Inaccurate or Incomplete Data:** The accuracy of the route profile is limited by potential errors in raw GPS data, and the interpolation of timestamps and altitudes introduces additional uncertainty.
- **Lack of Validation Data:** The consumption model has not been validated against real-world vehicle telemetry. This means that the predicted fuel and energy usage may not accurately reflect actual performance under varying conditions (e.g., different loads, weather, or driving styles).
- **Straight line approximation for detour:** Since we don't know which

Limitations of the MILP Solver

The Mixed-Integer Linear Programming (MILP) solver, while effective, has its own set of limitations:

- **Scalability Issues:** The node-based MILP model is computationally intensive and struggles to scale to large-scale fleet problems, even with node-reduction techniques.
- **Simplified Model Structure:** The solver simplifies certain real-world factors. For example, it only allows refueling at arrival nodes, and it approximates detour feasibility using a straight-line distance, which may not accurately reflect the actual road network.
- **Comparison with Existing Methods:** The acknowledges that other research, such as the work by Suzuki (2014)[10], has demonstrated more efficient approaches with tighter bounds and heuristics, which can solve problems with a larger number of refueling stops. Better approaches for different energies can be found in literature [4]

Implementation and Unverified Outputs

Practical implementation and the reliance on assumptions introduce further limitations:

- **Data Preprocessing Burden:** The model requires input data to be reformatted into a specific schema, which can be a time-consuming and error-prone process that might not align with real operational data.
- **Unverified Results:** The report's values for emissions, costs, and consumption are based on assumed parameters rather than validated data. For this reason, the report does not quote absolute numbers.
- **Incomplete Driver Constraints:** The model only partially accounts for driver-related regulations, such as rest breaks, which other research has shown can have a significant impact on optimal refueling plans[7].

Future Work:

- **Improving Data Quality:** Integrating real telematics data for consumption model calibration and linking to external APIs (e.g., OpenStreetMap) to improve the accuracy of station and detour data.
- **Enhancing the Solver:** Developing hybrid approaches that combine MILP with heuristics or exploring decomposition techniques to improve scalability for larger fleets.
- **Richer Modeling:** Through implementation of heuristics and tighter bounds. Ideally, this should make route -> journey conversion obsolete.
- **Validation:** Validating costs and emissions data with actual fleet operator datasets to ensure the model's outputs are reliable for decision-making.

5.2 Conclusion

As a prototype, the Greenfleet tool can be improved. The problem it tries to solve is not easy and more work needs to be done. Much complex approaches to this problem have been proposed. With this LP model ~480000 nodes were solved in a reasonable time with intended results and expected limitations.

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7 APPENDIX

7.1.1 Emission Factor by vehicle Type and Fuel

The emission factor for fuel consumption are reported in KgCO₂e/unit of fuel consumed.

Vehicle_type	B100	B7	Bio GNV	Electrique mix national (2020)	Electricity renewable	GNC blend	H2 low carbon	H2 low carbon (French Network)	H2 renewable	HVO
Porteur 19t	0.040	0.040	0.046	0.165	-	0.046	0.070	-	0.070	0.040
Porteur 26t	0.040	0.040	0.046	0.165	-	0.046	0.070	-	0.070	0.040
Rigide 12 Å 20t	0.045	0.045	0.051	0.169	-	0.051	-	0.062	0.062	0.045
Rigide 3,5 Å 7,5t	0.026	0.026	0.030	0.070	-	0.030	-	0.034	0.034	0.026
Tracteur Routier	0.045	0.045	0.051	0.129	0.129	0.051	0.089	-	0.089	0.045
VUL<3,5t - Compact	0.024	0.024	0.027	0.054	-	0.027	-	0.049	0.049	0.024
VUL<3,5t - Fourgon	0.031	0.031	0.036	0.082	-	0.036	-	0.060	0.060	0.031

Table 6 of emission factors.

7.1.2 Input parameters

This is a complete list of input parameters used for computing results. For reference only:

Category	Parameter	Value / Formula (units)	Where set	Notes
Run scope	Years simulated	range(2025, 2026) → 2025	main_sql.py	Controls which pump tables/prices are read.

	Fuels simulated	default EV	main_sql.py	Change fuel_types_enums to run others.
Vehicle	Cruise speed on route	60 km/h	main_sql.py	Used to estimate drive time and fill edge_time_map fallback.
	Detour speed	30 km/h	main_sql.py → VehicleParams	Used to turn radial detour distance into time.
	Tank/battery capacity	from catalog average; fallback 200	main_sql.py + vehicle_stats.py	Capacity by fuel for the vehicle type; 200 used if missing.
	Initial fuel	0.8 × tank	main_sql.py → VehicleParams	Starting level.
	Consumption for solver	distance-weighted avg from energy model	main_sql.py	Passed as fuel_consumption_rate.
Stations & detours	Station detour geometry	one-way off_km = beeline × 1.3	kampker_nodes.py	Round-trip distance in solver is 2 × off_km.
	Station filter	ignore stations with beeline > 100 km	main_sql.py	Overrides function's default 50 km.
	EV refuel time per unit	1 / charger_power_kw (h/kWh)	load_pump_data.py	Precomputed into time_per_unit_fuel.
	Non-EV refuel time per unit	1 / refuel_time (h/unit) if provided	load_pump_data.py	Stored as time_per_unit_fuel.
Energy / consumption	Default fallback rate	0.0741 (L/km, kg/km or kWh/km)	main_sql.py	Used if physics calc fails.
	Physics model params	g=9.81, PTRAG=gvw_kg (22 000 default), Cd=0.48, A=10 m ² , Crr=0.0055, Temp=median, Humid≈50%, Météo="Sec"	main_sql.py → consumption_nodes.py	Used in energy-balance and recuperation model, eco profile "standard".
	Speed for physics	from timestamps, else 60 km/h capped to [0.1,130]	main_sql.py	Builds per-edge speeds for the physics model.
Time constraints	Driver cost	€28 / h	main_sql.py	Used in objective and cost breakdown.
	Turn-around time	0 h (per intermediate node)	main_sql.py & solver.py	Included in total time as turn_around_h * max(0, n-2).
	Rest interval	4 h	main_sql.py	Used to compute mandatory rests post-solve.
	Max continuous driving	4 h	main_sql.py	Triggers 45-min mandatory rest when exceeded.
	Rest duration	45 min	main_sql.py	Duration of mandatory rest.
	Total time limit	est_drive_h × 50	main_sql.py	est_drive_h from timestamps, else total_km / speed.
	End-of-route reserve	0.3 × tank	main_sql.py → enforced in solver	Constraint fuel_before[n-1] ≥ min_fuel_end.
Solver mechanics	Fuel level bounds	lower 0.2 × tank ; upper 0.8 × tank for EV else 1.0 × tank	solver.py	Caps refuel headroom consistently for EV.
	Refuel time	EV: qty / power; non-EV: qty × time_per_unit	solver.py	Per-station timing constraint.

	Detour time & distance	$\text{time} = (2 \times \text{off_km}) / \text{detour_speed}$, $\text{distance} = 2 \times \text{off_km}$	solver.py	Applied when detour_bin=1.
	Objective (cost)	$\text{fuel_cost} + \text{driver_cost_per_hour} \times (\text{travel} + \text{detour} + \text{add_refuel} + \text{turnarounds})$	solver.py	Time components built from decision vars; separate breakdowns returned.
	Outputs (added)	refuel_time_cost, detour_time_cost, fuel_detour_cost	solver.py & main_sql.py	Exposed alongside totals for times, distances, quantities.

Table 7 Parameters used for simulation

7.1.3 Variation in Fuel prices over the years

The variation in fuel prices are reported in €/unit of fuel

	2025	2026	2027	2028	2029	2030	2031	2032	2033
B7	1.1217219	1.121722	1.534161	1.590988	1.647958	1.662572	0.167733	1.692242	1.707299
GNV	1.95445129	1.954451	1.918197	1.926795	1.935393	1.943991	1.952588	1.961186	1.969784
BIO GNV	2.18898544	2.188985	2.186745	2.235082	2.283763	2.332789	2.382158	2.451483	2.560719
EV	0.45	0.4553	0.4395	0.4377	0.4289	0.4395	0.4377	0.436	0.43773
B100	1.31967282	1.319673	1.319673	1.319673	1.319673	1.319673	1.319673	1.319673	1.319673
HVO	1.43844338	1.438443	1.438443	1.438443	1.438443	1.438443	1.438443	1.438443	1.438443
H2	12	11.5	11.01397	10.54847	10.10265	9.675676	9.266744	8.875096	8.875096

Table 8 Values used for a €/unit of fuel and projected from 2025-2032

7.1.4 Plot the journeys when used as a straight line segment

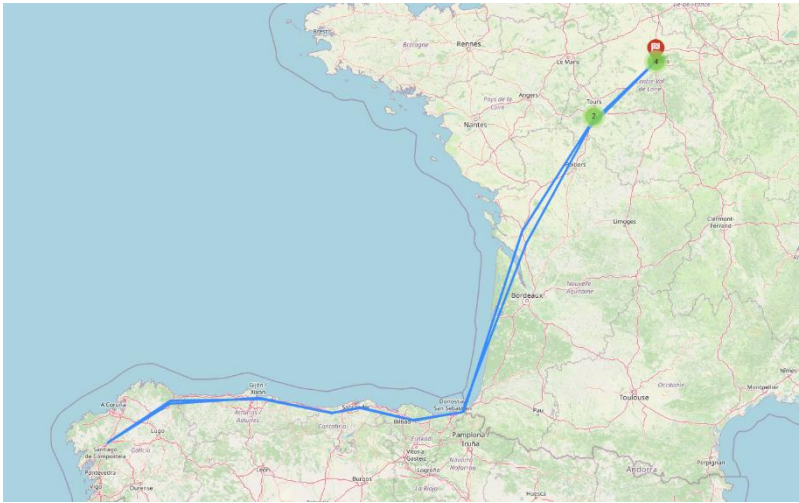


Figure 21: The straight line segment passes through the ocean, making projections unrealistic

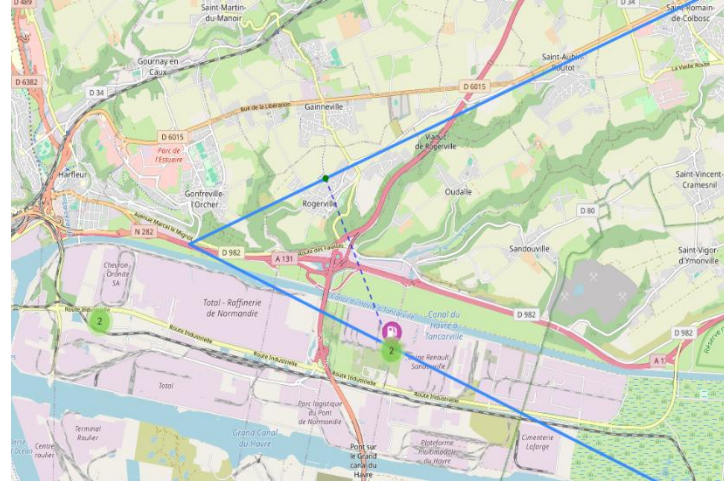


Figure 22: This vehicle goes clockwise. It skips closest pump and takes the same pump but mapped to the adjacent segment. This happens because the decision to refuel only happens at the start of the segment

Figure 15 and 16 uses model described in section 4.1 (The segment based MILP). The two problems mentioned; inaccurate routes and inaccurated refuel station assigned to segments can be visualized here. Using this approach we miss the nearby pumps as shown in Figure 16 and instead make a bigger detour which is not an optimal solution.

7.1.5 Modelling consumption

Modeling and determining fuel consumption is either determined by a “fuel map” which is a plot of RPM and fuel required. This is proprietary information and hard to get by.

Another method is to get the acceleration and deceleration of a vehicle, get the fuel consumption information and determine coefficients for a polynomial relation to a road load equation[15] .

An approximation was made to try to heuristically determine consumption based on scarce available data.

gradient_%	Speed km/h	EV kWh /100km	GNV Kg /100km	H2 kg /100km	B7 L /100km	B100 L/ 100km
0	30	53.16	13.39	3.57	14.66	14.66
0	60	60.91	15.34	4.09	16.80	16.80
0	90	88.65	22.33	5.95	24.44	24.44
0	120	128.52	32.37	8.62	35.44	35.44
2	30	194.19	48.91	13.03	53.54	53.54

2	60	185.71	46.77	12.46	51.21	51.21
2	90	213.45	53.76	14.32	58.85	58.85
2	120	253.32	63.80	17.00	69.85	69.85
5	30	405.32	102.08	27.20	111.76	111.76
5	60	372.56	93.83	25.00	102.72	102.72
5	90	400.294	100.815	26.862	110.37	110.37
5	120	440.166	110.857	29.537	121.36	121.36
8	30	615.479	155.01	41.302	169.7	169.7
8	60	558.537	140.669	37.481	154	154
8	90	586.273	147.654	39.342	161.65	161.65
8	120	626.144	157.696	42.018	172.64	172.64

Table 9 Consumption changing with gradient and speed for different energies

This work was prepared by Dan Trdak and used for computing consumption at each edge.

- **Air density & humidity handling:** Uses a simplified standard-atmosphere formula with water-vapor correction from relative humidity to get p for aero drag; altitude is taken from the node (if present), otherwise a default.
- **Weather → rolling resistance multiplier:** C_{rr} is scaled by a categorical weather factor (f_{meteo}): Sec = 1.0, Pluie = 1.15, Neige = 1.7, increasing rolling losses in wet/snow.

$$C_{rr}^{eff} = C_{rr} f_{meteo}$$

- **Speed- & temperature-dependent drivetrain “efficiency”:** A piecewise heuristic adjusts tank-to-wheel efficiency by fuel type, ambient temperature, and operating speed (penalized when very cold or crawling, boosted at moderate speeds, etc.). This is a compact proxy for powertrain maps.

$$\eta = \eta_{base} (fuel) \cdot f_{penalty}$$

$f_{penalty}$ is a multiplication factor for penalizing crawl (slow speed) and extreme temperatures

- **Eco-driving multipliers:** Three preset modes—standard, eco, eco_expert—scale speed profile, acceleration work, auxiliaries, and anticipative driving. These multiplicative factors reduce/increase energy terms without changing route geometry.
- **Auxiliary loads fixed at 2 kW (scaled by eco mode):** A constant auxiliary power is added as time-dependent energy; eco modes can down-rate it. This avoids modeling HVAC and ancillaries explicitly.
- **Aero yaw & wind treatment:** Apparent wind speed is computed from vehicle speed and a uniform wind with yaw angle; C_d is inflated by a yaw-dependent factor to reflect higher drag in crosswinds.

$$E_{aero,w} = \frac{1}{2} \rho C_d A_f v_{air}^2 d$$

- **Slope calculation preference order:** If a node has an explicit *slope* %, it's used directly; else slope is derived from altitude difference over edge distance—both feed gravitational work.

$$E_{slope,w} = m \cdot g \cdot \sin(\alpha) d.$$

- **Edge speed & time estimation:** For each edge, distance from cum_km; time from distance / average speed (with eco scaling). If timestamps exist, a “geometric” speed using measured Δt is recorded for diagnostics.
- **Weather/temperature inputs from pipeline (route-level heuristic):** Environment Specifications, temperature c is set to the **median** of journey points. temperature Celsius over the route; humidity is a simple heuristic linked to data presence. These feed the physics terms above.
- **Default parameter fallbacks:** If a vehicle/env field is missing, sensible defaults are used for mass (GVW), drag, frontal area, rolling resistance, weather, temperature, and humidity—ensuring the model always runs.

Default parameters given below were used for this report:

Parameter	Default Value	Unit	Description
g (m/s ²)	9.81	m/s ²	Gravitational acceleration
PTRA (kg)	22 000	kg	Gross vehicle weight (used as vehicle mass)
Aerodynamic drag coefficient	0.48	–	Aerodynamic drag coefficient CdC_dCd
A_f (m ²)	10	m ²	Frontal area of vehicle
C_{rr}	0.0055	–	Rolling resistance coefficient
Energy	Fuel Type	–	Selected energy carrier (B7, B100, GNV, H2, EV)
Temperature (°C)	15	°C	Ambient temperature
Humidity (%)	50	%	Relative humidity
Weather	"Sec"	categorical	Weather condition (Sec, Pluie, Neige)

Table 10 Default parameters for consumption used

7.1.6 Detailed TCO

The emissions are based on factors, the numbers are not accurate and hence not reported.

The table below is the aggregated output of the work done. The index is the immatriculation (unique vehicle id) concatenated with year and specific fuel. This table serves as an input for calculating Total TCO including CAPEX, insurance, depreciation, amortization etc. Once the total TCO is computed, we can optimize how the fleet has to be upgraded and create a roadmap for transition.

immatriculation	year	fuel_specific	total_cost	distance_km	fab_emission	fuel_emission	total_fuel_required	detour_km_sum
bq-265-be	2025	B100	506.8282	1301.13	58.76532	468.5482	384.0559	0

bq-265-be	2025	B7	430.804	1301.13	58.76532	1190.573	384.0559	0
bq-265-be	2025	Bio GNV	685.6268	1301.13	66.99246	292.0285	350.8027	0
bq-265-be	2025	Electricity nix national	767.1201	1301.13	167.9921	79.7053	1328.422	105.2489
bq-265-be	2025	Electricity renewable	767.1201	1301.13	167.9921	30.94558	1328.422	105.2489
bq-265-be	2025	GNC Blend	685.6268	1301.13	66.99246	461.06	350.8027	0
bq-265-be	2025	H2 low carbone	1114.87	1301.13	115.3903	305.6355	92.05889	5.549067
bq-265-be	2025	H2 renewable	1114.87	1301.13	115.3903	177.6737	92.05889	5.549067
bq-265-be	2032	B100	506.8283	1301.13	58.76532	468.5482	384.0559	0
bq-265-be	2032	B7	649.9156	1301.13	58.76532	1190.573	384.0559	0
bq-265-be	2032	Bio GNV	687.9893	1301.13	66.99246	213.6388	350.8027	0
bq-265-be	2032	Electricity nix national	748.5222	1301.13	167.9921	79.7053	1328.422	105.2489
bq-265-be	2032	Electricity renewable	748.5222	1301.13	167.9921	30.94558	1328.422	105.2489
bq-265-be	2032	GNC Blend	687.9893	1301.13	66.99246	461.06	350.8027	0
bq-265-be	2032	H2 low carbone	827.1951	1301.13	115.3903	305.6355	92.05889	5.549067
bq-265-be	2032	H2 renewable	827.1951	1301.13	115.3903	177.6737	92.05889	5.549067
bv-803-fd	2025	B100	195.4233	512.38	23.14156	180.6633	148.0847	0
bv-803-fd	2025	B7	166.1098	512.38	23.14156	459.0625	148.0847	0
bv-803-fd	2025	Bio GNV	264.3647	512.38	26.38138	112.6007	135.2629	0
bv-803-fd	2025	GNC Blend	264.3647	512.38	26.38138	177.776	135.2629	0
bv-803-fd	2025	H2 low carbone	431.8747	512.38	45.44026	117.6754	35.44439	3.401931
bv-803-fd	2025	H2 renewable	431.8747	512.38	45.44026	68.40767	35.44439	3.401931
bv-803-fd	2032	B100	195.4234	512.38	23.14156	180.6633	148.0847	0
bv-803-fd	2032	B7	250.5951	512.38	23.14156	459.0625	148.0847	0
bv-803-fd	2032	Bio GNV	265.2756	512.38	26.38138	82.37509	135.2629	0
bv-803-fd	2032	GNC Blend	265.2756	512.38	26.38138	177.776	135.2629	0

bv-803-fd	203 2	H2 low carbone	321.1 144	512.38	45.440 26	117.67 54	35.44439	3.40193 1
bv-803-fd	203 2	H2 renewable	321.1 144	512.38	45.440 26	68.407 67	35.44439	3.40193 1
cd-768-yp	202 5	B100	643.8 794	1798.5 2	81.229 86	595.24 82	487.9084	0
cd-768-yp	202 5	B7	547.2 975	1798.5 2	81.229 86	1512.5 16	487.9084	0
cd-768-yp	202 5	Bio GNV	872.6 453	1798.5 2	92.602 04	370.99 58	445.6631	1.03253 6
cd-768-yp	202 5	GNC Blend	872.6 453	1798.5 2	92.602 04	585.73 51	445.6631	1.03253 6
cd-768-yp	202 5	H2 low carbone	1437. 841	1798.5 2	159.50 12	388.45 52	117.0046	19.2804 8
cd-768-yp	202 5	H2 renewable	1437. 841	1798.5 2	159.50 12	225.81 88	117.0046	19.2804 8
cd-768-yp	203 2	B100	643.8 795	1798.5 2	81.229 86	595.24 82	487.9084	0
cd-768-yp	203 2	B7	825.6 59	1798.5 2	81.229 86	1512.5 16	487.9084	0
cd-768-yp	203 2	Bio GNV	875.6 729	1798.5 2	92.602 04	271.40 88	445.6631	1.03253 6
cd-768-yp	203 2	GNC Blend	875.6 729	1798.5 2	92.602 04	585.73 51	445.6631	1.03253 6
cd-768-yp	203 2	H2 low carbone	1072. 213	1798.5 2	159.50 12	388.45 52	117.0046	19.2804 8
cd-768-yp	203 2	H2 renewable	1072. 213	1798.5 2	159.50 12	225.81 88	117.0046	19.2804 8
cf-417-wa	202 5	B100	168.2 775	362.4	16.367 74	155.56 77	127.5145	0
cf-417-wa	202 5	B7	143.0 359	362.4	16.367 74	395.29 51	127.5145	0
cf-417-wa	202 5	Bio GNV	227.6 423	362.4	18.659 22	96.959 53	116.4738	0
cf-417-wa	202 5	GNC Blend	227.6 423	362.4	18.659 22	153.08 15	116.4738	0
cf-417-wa	202 5	H2 low carbone	371.8 826	362.4	32.139 33	101.65 96	30.62036	2.59203 2
cf-417-wa	202 5	H2 renewable	371.8 826	362.4	32.139 33	59.097 29	30.62036	2.59203 2
cf-417-wa	203 2	B100	168.2 775	362.4	16.367 74	155.56 77	127.5145	0
cf-417-wa	203 2	B7	215.7 855	362.4	16.367 74	395.29 51	127.5145	0
cf-417-wa	203 2	Bio GNV	228.4 268	362.4	18.659 22	70.932 54	116.4738	0
cf-417-wa	203 2	GNC Blend	228.4 268	362.4	18.659 22	153.08 15	116.4738	0
cf-417-wa	203 2	H2 low carbone	276.1 969	362.4	32.139 33	101.65 96	30.62036	2.59203 2

cf-417-wa	203 2	H2 renewable	276.1 969	362.4	32.139 33	59.097 29	30.62036	2.59203 2
cq-596-pm	202 5	B100	952.7 156	2319.8 2	104.77 43	833.42 44	683.1347	37.5685 9
cq-596-pm	202 5	B7	767.4 936	2319.8 2	104.77 43	2117.7 18	683.1347	0.49544 3
cq-596-pm	202 5	Bio GNV	1232. 752	2319.8 2	119.44 27	519.44 21	623.9859	5.62600 2
cq-596-pm	202 5	GNC Blend	1232. 752	2319.8 2	119.44 27	820.10 47	623.9859	5.62600 2
cq-596-pm	202 5	H2 low carbone	2264. 54	2319.8 2	205.73 25	541.31 58	163.0469	178.887 8
cq-596-pm	202 5	H2 renewable	2264. 54	2319.8 2	205.73 25	314.68 06	163.0469	178.887 8
cq-596-pm	203 2	B7	1157. 345	2319.8 2	104.77 43	2117.7 18	683.1347	0.49544 3
cq-596-pm	203 2	Bio GNV	1237. 109	2319.8 2	119.44 27	380.00 74	623.9859	5.62600 2
cq-596-pm	203 2	GNC Blend	1237. 109	2319.8 2	119.44 27	820.10 47	623.9859	5.62600 2
cq-596-pm	203 2	H2 low carbone	1755. 034	2319.8 2	205.73 25	541.31 58	163.0469	178.887 8
cq-596-pm	203 2	H2 renewable	1755. 034	2319.8 2	205.73 25	314.68 06	163.0469	178.887 8

Table 11 Detailed consumption TCO for vehicles over given routes

For simulation the prices of specific fuel like GNV/Bio GNV is kept constant and no distinction is made yet.